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28 October 1977
SUPERSEDING
MIL-R-28707B(EC)
1 November 1973
(See 6.7)

MILITARY SPECIFICATION

RADIO SET AN/URC-35( )

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Radio Set AN/URC-35( ) and associated accessories. This equipment is a digitally tuned transceiver, intended for general purpose voice and CW communications, and may be used for surface ships, fixed shore installations, in amphibious landing craft, in vehicles, and for use aboard ship as an emergency radio. The equipment operates in the 2.0 to 30.0 megahertz (MHz) frequency range, and provides transmission and reception of single sideband (selectable upper and lower sideband), CW and AM (compatible) signals.

1.2 Operational requirements. Each unit shall give reliable performance within the limits specified herein, under conditions of intermittent or continuous operation during exposure to the adverse conditions normally encountered in Naval ship and shore communications service. Mechanical and electrical ruggedness is of paramount importance for assurance of reliable performance without failure.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.i. (High-impact), Shipboard Machinery, Equipment And Systems, Requirements For (NAVY)
MIL-C-915	Cable And Cord Electrical, For Shipboard Use, General Specification For
MIL-M-10304	Meter, Electrical Indicating, Panel Type, Ruggedized, General Specification For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-W-16878	Wire, Electrical, Insulated, High Temperature (NAVY)
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-F-19207	Fuseholder, Extractor Post Type, Blown Fuse Indicating And Nonindicating, General Specification For
MIL-C-22954	Control, Radio Set C-1138( )/UR
MIL-A-23547	Antenna Coupler CU-937( )/UR
MIL-M-38510	Microcircuits, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type 1-Environmental and Type 2-Internally Excited)
MIL-STD-189	Rack, Electrical Equipment, 19-INCH And Associated Panels
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-883	Test Methods And Procedures For Microelectronics
MIL-STD-965	Parts Control Program

DRAWINGS

MILITARY

NAVAL SEA SYSTEMS COMMAND

RE-C2687412	Electrical Power Cable Assembly CX-1846( )/U
RE-F2688036	Handset Assembly H-169( )/U
RE-B2696065	Installation Control Drawings, Data List, For AN/URC-35( )

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

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3. REQUIREMENTS

3.1 General. Equipments shall be in accordance with the provisions of MIL-E-16400, to the extent specified herein.

3.1.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3). The sample(s) shall be suitable for complete evaluation of mechanical and electrical form, design and performance. It shall be of final mechanical and electrical form, employ standard or nonstandard (when approved by the command or agency concerned) parts, and be completely representative of final equipment.

3.1.2 Parts, materials and processes. Parts, materials and processes shall be in accordance with MIL-E-16400, except as otherwise specified herein. Particular attention shall be paid to the acceptable and unacceptable materials of MIL-E-16400.

3.1.2.1 Selection of parts, materials and processes. Selection of parts, materials and processes shall be in accordance with MIL-E-16400. Standard parts are as indicated in MIL-E-16400.

3.1.2.2 Approval of nonstandard parts. Requests for approval for the use of nonstandard parts shall be submitted in strict accordance with the requirements of MIL-STD-965.

3.1.2.2.1 Prescreening of approval requests. Prescreening of nonstandard parts approval requests shall be accomplished in accordance with MIL-STD-965.

3.1.2.2.2 All nonstandard parts shall have been documented in accordance with the requirements of MIL-STD-965, and approved, prior to the conduct of first article tests required by 4.3.

3.1.2.2.3 All approved nonstandard parts shall be subjected to quality conformance inspection, in accordance with the pertinent component specification, as specified in MIL-E-16400. Inspection shall be as specified by the procuring activity upon submission, otherwise Group A only in accordance with the pertinent component specification.

3.1.2.2.4 If any Government furnished equipment (GFE) is provided, it may include nonstandard parts. The equipments furnished in accordance with this specification shall employ standard parts or approved nonstandard parts. Nonstandard part approvals previously granted under earlier contracts for equipments represented by the GFE shall not be construed as being extended to equipments furnished in accordance with this specification. Nonstandard parts cited in Government furnished drawings shall not be construed as being approved for use in equipments furnished in accordance with this specification.

3.1.2.3 Individual selection. The performance of the equipment shall not be dependent on the selection of individual parts, unless such selection is specifically approved by the procuring activity. If such individual part selection is approved, sufficient information and data shall be included as is necessary in all applicable technical publications and engineering drawings to enable designated repair activities to perform such selection.

3.1.2.4 Electron tubes. All electron tubes shall be selected from among approved types in accordance with MIL-E-16400. Electron tubes shall be limited to applications corresponding to circuits of the GFE.

3.1.2.5 Microelectronic devices. Selection and application of microelectronic devices shall comply with Requirement 64 of MIL-STD-454 requiring MIL-M-38510 Class B devices. When deviations to these requirements are necessitated due to availability or cost considerations, hermetically sealed devices inspected and tested to the requirements of MIL-STD-883 Class B may be used.

3.1.2.6 Semiconductor devices. All semiconductor devices shall be made of silicon. This was not a requirement for early models of this equipment, and may require redesign.

3.1.3 Production equipment. Production equipments shall be identical to the equipment that has been subjected to and passed the first article examinations and tests (see 3.1.1 and 4.3).

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3.2 Government furnished material (GFM). Government furnished material, if any, shall be as specified (see 6.2).

3.3 Interchangeability. Interchangeability requirements, if any, shall be as specified in the contract or order (see 6.2).

3.4 Composition. The Radio Set AN/URC-35( ) shall consist of one each of the following:

a. Receiver-Transmitter RT-618( )/URC
b. Radio Frequency Amplifier AM-3007( )/URT (with 115 VAC power supply provided and installed)
c. Shock and Vibration Mounting Assembly MT-3761( )/UR (see 3.8.1)
d. Antenna Coupler CU-937( )/UR, in accordance with MIL-A-23547
e. Interconnection Box J-2940( )/URC-35
f. Handset H-169( )/U, in accordance with RE-F2688036, including cord and plug assembly CX-1846( )/U, in accordance with RE-C2687412
g. Set of cables for interconnection of Receiver-Transmitter, RF Amplifier and Interconnection Box
h. Set of mating connectors for all receptacles for which interconnection cables of (g) are not furnished
i. Set of mounting brackets

3.4.1 Accessories. In addition to the basic radio set of 3.4, the following accessory items shall be furnished in the quantities as specified (see 6.2):

a. Radio Remote Control Unit C-9044( )/URC-35
b. Antenna AT-1047( )/U (16 foot sectionalized, gray color Columbia Products, Inc. model 120A with feedthrough base, or equivalent approved by the command or agency concerned)
c. Equipment plug-in assembly extension cables, as follows:
    1. RF Amplifier Assembly Cable (2 each)
    2. Frequency Standard Assembly Cable (1 each)
    3. Translator/Synthesizer Assembly Cables (2 each)
    4. Receiver IF/Audio Cable (1 each)
    5. Transmit IF Assembly Cable (1 each)
    6. Mode Selector Assembly Cable (2 each)
    7. Transmit Audio/RF Amplifier APC/PPC Assembly Cable (1 each)
    8. RF Amplifier Power Supply Assembly Cable (1 each)
    9. RF Amplifier DC/DC Converter Assembly Cables (2 each)
    10. RF Amplifier APC/PPC Assembly Cable (2 each)
d. Set of printed circuit board (PCB) extender cards, one for each PCB in the equipment

3.5 Frequency range. The frequency (tuning) range of the equipment shall be from 2.0 to 30.0 MHz. Each equipment shall be capable of adjustment to any frequency from 2.0 to 29.9999 MHz in 100 hertz (Hz) increments. In addition, in the recieve modes, continuous coverage shall be provided by a vernier frequency control capable of tuning between one kilohertz (kHz) increments (see 3.16.2.1.1.2).

3.5.1 Method of tuning. Frequency selection over the complete frequency range of the equipment shall be accomplished in increments of 100 Hz, where changing the setting of the digital tuning knobs is all that is required to tune the equipment (see 3.15.1.6). Frequency tuning of the RF amplifier shall be accomplished automatically when the digital tuning knobs of the receiver-transmitter are switched.

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3.6 Primary power source. Each equipment shall be capable of operating and providing rated performance from a primary power source as follows (optionally selectable):

a. 115 volts AC (VAC) ±10 percent, 48 to 420 Hz, single phase
b. 24 to 30 volts DC (VDC) external input source

This capability shall occur with no degradation of performance by power line variations (relative to the nominal) of ±10 percent (AC) or ±2.5 VDC, or above specified power line frequency variations, with either of these variations acting separately or in combination. The primary of the AC power transformer shall be equipped with taps for 90, 95, 100, 105, and 110 percent of the nominal. The equipment shall be delivered and tested with connection to the normal 115 volt tap. The total power consumption at 115 volts, 60 or 400 Hz, shall not exceed 500 watts.

3.6.1 The equipment shall meet the performance limits of operation under the conditions of transient voltage, transient frequency and power interruption as specified in MIL-E-16400 for shipboard requirements. The equipment shall include provisions to protect against transients generated when the tuning motors energize or deenergize.

3.6.2 Electrical overload protection. Electrical overload protection shall be in accordance with MIL-E-16400. Each line of the alternating current (AC) power input shall be protected by a circuit breaker or fuse.

3.6.3 In a system installation, power shall enter the equipment's interconnection box and be distributed by way of the main interconnection cabling to the equipment's AM-3007( )/URT RF Amplifier and RT-618( )/URC Receiver-Transmitter units (see 3.4).

3.6.4 Unless otherwise specified in the contract or order, the AN/URC-35( ), as delivered, shall contain a power supply for operation of the equipment from a 115 volts alternating current (VAC) ±10 percent, 48 to 420 Hz, singlephase, power source.

3.6.4.1 The bleeder and filter networks for the power supply shall provide regulation to preclude visible variations in panel lamp incandescence when voice intelligence is applied to the equipment.

3.6.4.2 The filament voltage of electron tubes, with the exception of the AM-3007( )/URT final amplifier, shall be held at 6.3 volts ±5 percent for all primary power source voltage variations described in 3.6, and for all environmental conditions in which operation is required.

3.6.5 Suitable protection shall be provided to prevent equipment damage due to application of reverse polarity direct current (DC) power.

3.7 Operational requirements.

3.7.1 Type of emission. Each equipment shall provide for the transmission and reception of the following types of emission:

a. A1 Telegraphy without the use of a modulating audio frequency (CW)
b. A3a Telephony, single sideband (SSB), reduced carrier
c. A3h Telephony, single sideband (SSB), full carrier
d. A3j Telephony, single sideband (SSB), suppressed carrier
e. A7j Single or multichannel voice frequency telegraphy (RATT), reception only

3.7.2 Modes of operation. The equipment shall be capable of transmitting and receiving the emissions described in 3.7.1 by selection of one of the following modes of operation:

a. LSB - Lower sideband
b. USB - Upper sideband
c. AM - Amplitude modulation (compatible)
d. CW - Continuous wave telegraphy

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3.8 Equipment arrangement. The normal installation (stacking) arrangement for the AN/URC-35( ) shall be as follows, starting from the bottom mounting surface:

a. Shock and Vibration Mounting Assembly MT-3761( )/UR
b. Receiver-Transmitter RT-618( )/URC
c. RF Amplifier AM-3007( )/URT

3.8.1 Shock and vibration mounting assembly, MT-3761( )/UR. Each AN/URC-35( ) shall be delivered with the necessary brackets and hardware required to attach the units to the mounting assembly.

3.8.2 The Antenna Coupler CU-937( )/UR shall be installed as indicated in MIL-A-23547.

3.8.3 Changes from prior AN/URC-35( ) primary power requirements. Previous AN/URC-35( ) Radio Sets were required to operate from an internally installed 24 V battery. This requirement no longer exists. GFE and documentation may include provisions for battery operation. Such provisions are to be deleted from equipment delivered in accordance with this specification.

3.9 Radio set system. The contractor shall be responsible for demonstrating complete system integration and system operation, for the equipment to be delivered in accordance with the requirements of this specification. Equipment provisions for external cabling to the standard Navy shipboard radio remote control systems shall be required. Standard Navy cables and related information are contained in MIL-C-915. Mating connectors for all external cabling requiring installing activity manufacture shall be furnished with each equipment. Interconnection cabling, plugs and receptacles for the interconnection of the RT-618( )/URC, AM-3007( )/URT and interconnection box units shall be supplied with each AN/URC-35( ) equipment. Additional information pertaining to installation of the equipment may be found in the Installation Control Drawings, RE-B2696065.

3.9.1 Radio remote control system. The equipment shall contain compatible provisions for remote control of the Radio set in the standard Navy radio remote control system (not frequency control) using a maximum of three Radio Set Control Units C-1138( )/UR in accordance with MIL-C-22954. Continuous simultaneous key-down operation from only two C-1138( )/UR units shall be required. Simultaneous connection and use of the C-9044( )/URC-35 remote control unit and the standard Navy C-1138( )/UR control units will not be required. The equipment shall be capable of operation remotely through the use of one C-9044( )/URC-35 remote control unit. Start-stop provisions are not required with use of C-1138( )/UR control units.

3.10 Dimensions and weights. The maximum dimensions and weights of the equipment shall not exceed those shown in TABLE I.

TABLE I. Dimensions and weights.

Unit	Width Inches (cm)	Depth 1/ Inches (cm)	Height Inches (cm)	Weight Pounds (kg)
Receiver-Transmitter	17-3/8 (44.13)	18-1/2 (47.0)	7 (17.75)	75 (34.0)
RF Amplifier (with 115 VAC power supply installed)	17-3/8 (44.13)	16-1/2 (42.5)	7 (17.75)	80 (36.4)
Shock/Vibration mount assembly	19-3/4 (51.0)	16-3/4 (43.2)	3-5/8 2/ (9.35)	17 (7.52)
Antenna coupler	12 (30.5)	22 (length) (56.0)	12 (30.5)	30 (13.6)
Remote control unit	8 (20.3)	6 (15.2)	6 (15.2)	6 (2.72)
Interconnection box	11-1/4 (28.2)	8 (20.3)	5-1/2 (14.0)	10 (4.54)

1/ Exclusive of rear connectors
2/ Maximum unloaded height of bedplate

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3.11 Environmental service conditions. The equipment shall be capable of continuous, reliable operation, within the performance limits specified herein, under those environmental conditions of MIL-E-16400 listed herein. The equipment shall operate within tolerances throughout the corresponding test specified in section 4 without alignment or adjustment, other than external controls normally employed for operation of the equipment.

a. Operating temperature range (Range 2(modified)): -28°C to +50°C
b. Non-operating temperature range
c. Humidity (0 to 100 percent)
d. Salt fog (spray) (unsheltered)
e. Shock
f. Vibration
g. Inclination (60 degrees)

3.12 Orientation. The equipment shall be capable of storage in any orientation.

3.13 Reliability.

3.13.1 Quantitative reliability requirement. The specified mean-time-between-failures (MTBF) (θ_o as defined by MIL-STD-781) for this equipment shall be 500 hours.

3.13.2 Environmental burn-in. All equipments shall be subjected to an environmental burn-in for a cumulative operating time of at least 24 hours. In receive mode, the equipment shall be operated using a received RF signal of one microvolt at any frequency within the specified frequency range. Audio output shall be set at maximum. In transmit mode, the equipment shall be operated at a nominal 100 watts output at any frequency within the specified frequency range. No adjustments of the equipment, other than operating adjustments, shall be allowed. A five minute transmit and a five minute receive duty cycle shall be used.

3.14 Basic conditions. The following basic conditions shall apply as pertinent to the performance requirements specified herein, except as modified by a particular test or test procedure contained in this specification. These conditions are specifically applicable to the measurement of spurious response and oscillator radiation characteristics in the receiver section of the receiver-transmitter; measurement of the output power, spurious responses, intermodulation distortion, bandwidth and carrier suppression in the transmitter section of the receiver-transmitter; and the measurement of the power output, harmonic output and distortion in the RF amplifier unit.

3.14.1 Standard test conditions.

3.14.1.1 Primary power. The primary power test condition shall be 115 VAC, 60 Hz, single phase.

3.14.1.2 Ambient temperature. The ambient temperature shall be between 20° and 30° C.

3.14.1.3 Relative humidity. The relative humidity shall be less than 95 percent.

3.14.1.4 External/internal frequency standard switch. The FREQ STD switch shall be adjusted to its INT/COMP position.

3.14.1.5 Internal frequency standard. The internal frequency standard adjustment control shall be adjusted as required to ensure correct setting of the output frequency.

3.14.1.6 Transmitter RF output. Transmitter RF output shall be measured with a 50 ohm resistive load connected to the RF output terminal through a suitable transmission line.

3.14.1.7 RF amplifier RF input. All RF amplifier RF inputs shall be applied to the unit from a standard source impedance.

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3.14.1.8 RF amplifier bias. The RF amplifier bias shall be adjusted to ensure that the correct operating point has been set for the tubes.

3.14.1.9 Audio frequency (AF) outputs. Unless otherwise specified herein, all AF outputs shall be measured with a 600 ohm nonreactive load connected across the output terminals.

3.14.1.10 RF input to receiver. All RF inputs to the receiver shall be applied to the antenna input terminals through a standard dummy antenna.

3.14.1.11 Receiver phone level control. The receiver phone level control shall be adjusted as required herein for the determination of performance of the receiver.

3.14.1.12 Receiver beat frequency oscillator (BFO) control. Normally, the BFO control shall be adjusted at its zero or mid-scale position, but may otherwise be adjusted, as required herein, to determine the performance of the receiver.

3.14.1.13 *Audio frequency remote inputs. Unless otherwise specified herein, AF inputs will* be applied to the remote input terminals that interface with the Navy shipboard remote system.

3.14.1.14 Audio frequency local input. Local input performance will be demonstrated using the H-169/U Handset or an equivalent device.

3.15 Mechanical design and construction. Mechanical design and construction furnished in accordance with this specification shall conform to all applicable requirements of MIL-E-16400, to the extent of being compatible with the GFE.

3.15.1 General requirements.

3.15.1.1 Construction. Construction shall conform to the requirements of MIL-E-16400. The receiver-transmitter and RF amplifier shall consist basically of a frame, a front panel, a case and a group of replaceable plug-in assemblies. Retractable frame slides shall be provided on the receiver-transmitter and RF amplifier units that will permit the frame, when withdrawn from the case, to be oriented ±90 degrees from its normal horizontal position. Locks shall be provided to maintain the frame in the desired position. The frame shall include the power supply (receiver-transmitter), receptacles for insertion of the plug-in assemblies, and required control functions. The front panel shall be securely mounted to the frame, with two handles provided, one mounted on each side of the front panel and each capable of supporting the entire weight of the unit.

3.15.1.1.1 The baseplate of the translator synthesizer shall be a rectangular piece, eliminating the protruding, securing flanges of the earlier versions of this assembly.

3.15.1.2 Case. The case shall be made of aluminum. The case shall be of an extruded construction, the same as that of the GFE. A retractable cable and a frame mating receptacle shall be provided to enable operation of the equipment when the frame is withdrawn from the case and oriented ±90 degrees from its normal horizontal position. Additional accessory cables shall not be required for operation when the equipment is withdrawn from the case. The retractable cable shall be installed such that it does not jam when the frame is inserted in or withdrawn from the case.

3.15.1.2.1 Rack mounting. The equipment shall be designed to permit standard rack mounting, in accordance with MIL-STD-189, when required. The equipment shall be designed and constructed so that when mounted in a standard rack, and not subject to shock, no other support or reinforcing members will be required to preclude deformation of any part of the equipment. The equipment's technical manual shall also include a sample fabrication drawing for the manufacture, by an installing activity, of suitable mounting brackets that will permit mounting of the units in a standard mounting rack, providing the notching conforms to MIL-STD-189.

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3.15.1.3 Enclosure. The enclosure for the equipment shall be in accordance with MIL-E-16400. The degree of enclosure of the equipment shall be splashproof.

3.15.1.3.1 External connections. External connections shall be the same as in the GFE.

3.15.1.4 Bonding and grounding. The equipment shall meet the requirements of MIL-E-16400, except as otherwise specified herein.

3.15.1.4.1 Case bonding. Case bonding and grounding shall be accomplished by preparing mounting surfaces to provide metal contact between the case and mounting bracket, and between the mounting bracket and shock mount. The area around each threaded insert of the case used to secure the mounting bracket to the case shall be free of paint and chemically treated to preserve the metal and provide a ground. Mounting brackets and the shock mount used with the equipment shall be similarly treated to insure bonding and grounding.

3.15.1.4.2 Grounding stud. The stud shall be a pressed-in captive insert made integral with the case and constructed of austenitic corrosion-resistant steel. The captive insert shall be flush with the outside surface of the case and shall contain a blind threaded hole to prevent the entrance of moisture into the case. A 25.4 mm (one-inch) diameter area around the threaded insert shall be free of paint and chemically treated to preserve the metal and provide a ground.

3.15.1.5 Corrosion protection, corrosion resisting treatments and painting. Corrosion protection and corrosion resisting treatments shall be in accordance with MIL-E-16400.

3.15.1.5.1 Painting. Painting shall be in accordance with MIL-E-16400, except that the interior surfaces of each metal equipment enclosure shall be painted a dull black to improve thermal efficiency.

3.15.1.6 Frequency selection system. A digital frequency selection system shall be provided. Manually operated knobs on the front panel shall effectively control the frequency and the display of frequency to which the equipment is tuned. Separate controls shall be provided for the 10 MHz, 1 MHz, 100 kHz, 10 kHz, 1 kHz and 100 Hz frequency selection, and each control shall cover its range in less than one rotation of the knob.

3.15.1.6.1 The backlash in the mechanical tuning arrangement shall not abrogate the requirements for frequency accuracy and stability. End stops shall not be provided, except for the 1 and 10 MHz and 100 Hz controls. Each digit selected shall include mechanical detent provisions.

3.15.1.6.2 The sequence of positioning the frequency selector knobs or the manner or speed in which the knobs are rotated shall not affect any of the performance requirements specified herein. It shall not be necessary to position any frequency selector knob more than once in obtaining the desired frequency.

3.15.1.7 Fastener hardware. Fastener hardware shall meet the requirements of MIL-E-16400, except as otherwise specified herein. All screws, nuts, bolts, and other fastener hardware shall be of austenitic corrosion-resistant steel, except in those applications requiring material capable of withstanding high tensile stresses. Captive-type bolts shall be used on equipment front panels and electronic plug-in assemblies. Cadmium plating is not acceptable on the captive-type bolts used on the equipment front panels and electronic plug-in assemblies.

3.15.1.8 Controls, indicators and panel layouts. Controls, indicators and panel layouts, including lights, receptacles, switches, meters (including functions) and fuseholders, shall be the same as furnished in the GFE, unless otherwise specified herein. A dimmer control for the equipment panel lights is not required.

3.15.1.8.1 Mode selector switch. The mode selector switch shall function in a manner which requires a deliberate action in addition to rotation to accomplish power turnoff, which would result in shutdown of the frequency standard. Design of this control shall be included as part of the panel layout, if required.

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3.15.1.9 Electrical and radio frequency (RF) connectors. Electrical and radio frequency connectors shall be provided in accordance with and meet the requirements of MIL-E-16400. All non-RF connectors shall be of the solder-lug type; crimped pin types shall not be used. Mating connectors for non-RF connectors shall be provided with cable clamps.

3.15.1.9.1 RF input and output connectors. The coaxial radio frequency connectors for the external frequency standard input and the antenna input/output shall be Type N. The coaxial radio frequency connector for the internal frequency standard output shall be Type BNC. The receptacles shall be mounted at the rear of the equipment cases and shall be suitably labeled. The center pin of the type N connectors shall be made captive in all directions.

3.15.1.9.2 Assembly (module) connectors. RF inserts, if used, shall be constrained from nonperpendicular movement to ensure proper mating.

3.15.1.10 Time meter. A time meter is not required.

3.15.1.11 Locking devices. Locking devices on external operating controls are not required unless otherwise specified herein.

3.15.1.12 Panel lights. Panel lights shall be easily replaceable from the front panel without requiring removal of the frame from its case.

3.15.1.13 Motors. Motors used in the equipment shall comply with the requirements of MIL-E-16400, to the extent of compatibility with the GFE.

3.15.1.14 Fuses. Fuses shall be in accordance with MIL-E-16400, except that extra fuses are not required. Fuses shall be mounted on the front panel in blown fuse indicating fuseholders which conform to MIL-F-19207. Spare fuseholders are not required.

3.15.1.15 Filters. Filters used in the sideband selection process shall be electro-mechanical or crystal filter type.

3.15.1.16 Phone jack. A phone jack shall be provided on the front panel to enable the use of a headset in place of a speaker for full operation of the equipment.

3.15.1.17 Head telephones, hand keys and plugs. Head telephones, hand keys (CW) and telephone plugs are not required to be furnished by the contractor.

3.15.1.18 Assembly identification. All assemblies to which the interchangeability requirements apply (see 3.3) shall be provided with the appropriate assigned National Stock Number, Manufacturer's Part Number (including revision indicator), Federal Supply Code Number, and Serial Number. This information shall be inscribed on the baseplate of the assembly.

3.15.1.18.1 Dust covers. The dust covers of all assemblies employing printed circuit boards shall indicate the location of all test points and all adjustments accessible through the dust cover.

3.15.1.19 Airborne and structureborne noise. The airborne and structureborne noise requirements of MIL-E-16400 are not applicable.

3.15.1.20 Accessibility. In addition to the requirements of MIL-E-16400, the circuit mounting boards, components, and tubes shall be easily accessible for maintenance, repair or replacement. This requirement shall be satisfied to the extent of being compatible with the GFE. Components which require adjustment prior to start of operation shall be accessible without having to remove dust covers or other coverings.

3.15.1.21 Test provisions. Test points shall be so provided and located as to permit the isolation of malfunctioning parts or subassemblies within any major unit or assembly, without the unsoldering of any interconnection or wiring lead. Test points provided shall be identical with those extant in the GFE.

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3.15.1.21.1 AM-3007( )/URT test points. Test points shall also be provided in the AM-3007( )/URT to measure the grid inputs to the tubes and at the input to the RF bridge circuit.

3.15.1.22 Circuit alignment. Resonant circuits shall be provided with adjustable trimmers, or equally effective means, to permit accurate alignment. The trimmers shall provide sufficient means of adjustment to compensate for the effects of normal aging of electron tubes or other parts. If shrinkable tubing is used on variable components, it shall permit necessary adjustments to be made without having to replace the tubing.

3.15.1.22.1 Composite bandpass filters may be furnished as sealed units without trimmers, provided the input and output sections do not require adjustment.

3.15.1.23 Factory adjustment controls. The design of the equipment shall not include factory or sealed adjustment controls, unless specifically approved by the procuring activity. Where these controls are approved, a complete description and instructions for adjustment of such controls shall be included in the associated technical publications.

3.15.1.24 Shock, vibration, and inclination.

3.15.1.24.1 Shock. The equipment shall be capable of withstanding the shock test for Grade A, deck mounted, Class II, lightweight, Type A equipment as specified in MIL-S-901.

3.15.1.24.2 Vibration. The equipment shall be capable of withstanding the Type I vibration test of MIL-STD-167-1. During vibration testing, the equipment shall conform to the requirements of 3.16.1.4.2. The nameplate requirements of MIL-STD-167-1 are not applicable.

3.15.1.24.3 Inclination. The equipment shall conform to the inclination requirements of 3.11 and MIL-E-16400.

3.15.1.24.4 Shock and vibration mount distortion. The shock and vibration mounting assembly shall show no deformation, bottoming or leaning on completion of the shock, vibration or inclination tests.

3.15.1.25 Printed wiring. The fabrication of printed wiring boards and the insertion of parts and assemblies thereon shall conform to the requirements of MIL-E-16400, to the extent of being compatible with the GFE.

3.15.1.26 Soldering. Soldering shall conform to the requirements of MIL-E-16400.

3.15.1.27 Welding. Welding shall conform to the requirements of MIL-E-16400.

3.15.1.28 Attachment of wires and leads. Attachment of wires and leads shall conform to the requirements of MIL-E-16400.

3.15.1.29 Internal wiring and cabling. Internal wiring and cabling shall conform to the requirements of MIL-E-16400. All wiring shall be TFE insulated. Hook-up wire shall conform to MIL-W-16878, Type E.

3.15.1.30 Conductor identification. Identification of conductors shall conform to the requirements of MIL-E-16400. The method used for identification shall be consistent throughout the equipment.

3.15.1.31 Identification and marking. Identification and marking of all units, assemblies, electrical and moving mechanical parts, and any mechanical parts required to be removed for normal maintenance, shall be in accordance with MIL-E-16400. Hertz or Hz shall be used for all indications of frequency on hardware and associated software. Reference designators shall be the same as used in the GFE to the maximum extent practicable.

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3.15.1.32 Part replacement. Part replacement shall conform to the requirements of MIL-E-16400.

3.15.1.33 Quartz crystal units. Quartz crystal units shall conform to the requirements of MIL-E-16400.

3.15.1.34 Gears and cams. Gears and cams shall conform to the requirements of MIL-E-16400, except that only metal gears shall be used.

3.15.1.35 IF/Audio Assembly adjustment. All maintenance adjustments on the IF/Audio Assembly shall be accessible from the top without removal of the dust cover.

3.15.1.36 Extender cables. Extender cables for each assembly listed in 3.4.1c shall be furnished as required (see 6.2). Each completed cable shall be 609.6 mm (24 inches) long. These cables will be used to assist in maintaining the equipment.

3.15.1.37 Frequency standard assembly construction (maintenance considerations). Construction of the frequency standard assembly shall permit maintenance or replacement, or both, of: a) the oscillator and oven control circuits and, b) the oven case as separate plug-in subassemblies. The oven case subassembly components (except the heater winding) shall be readily removable for replacement. No potting or molding processes shall be employed to secure replaceable components to the oven case.

3.15.1.38 Thermal design. Thermal design shall be in accordance with MIL-E-16400 to the extent of compatibility with the GFE. A thermal survey shall be performed.

3.15.2 Antenna Coupler CU-937( )/UR. The CU-937( )/UR shall meet the construction requirements of MIL-A-23547.

3.15.3 Shock and vibration mounting assembly MT-3761( )/UR. The shock and vibration mounting assembly shall comply with the requirements of MIL-E-16400. The mount when attached to the AN/URC-35( ) shall enable the equipment to comply with the requirements of 3.15.1.24.

3.15.3.1 The assembly shall be streamlined to the extent that it shall have no sharp corners or edges. All corners shall be rounded.

3.15.4 Remote control unit C-9044( )/URC-35.

3.15.4.1 The C-9044( )/URC-35 shall be capable of surface mounting in a location removed from the AN/URC-35 equipment.

3.15.4.2 The C-9044( )/URC-35 shall be provided with the following front panel controls:

a. Power switch
b. Volume control
c. Power indicator
d. Phone-speaker selector switch
e. H-169( )/U handset jack

3.15.5 Interconnection box J-2940( )/URC-35.

3.15.5.1 The J-2940( )/URC-35 shall be capable of being mounted on a flat vertical surface in a location removed from the AN/URC-35. The interconnecting cable to be furnished (see 3.4 g) shall be 32.81 meters (ten feet) long.

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3.15.5.2 The following hardware shall be provided as part of the J-2940( )/URC-35:

a. Watertight access cover
b. Connector to mate with interconnecting cable
c. Stuffing tube, size 5
d. Terminal boards
e. Ground strap

3.15.6 Mounting brackets.

3.15.6.1 The brackets shall be streamlined to the extent that they shall have no sharp corners or edges. All corners shall be rounded.

3.16 Detailed performance.

3.16.1 General (system) requirements. The following requirements are for operation using a dummy load in lieu of the antenna coupler.

3.16.1.1 Power output. The equipment shall be capable of transmitting 100 watts PEP (minimum) in the SSB mode (USB or LSB), 50 watts (minimum) in CW mode, and 25 watts carrier (minimum) in the AM mode, into a 50 ohm resistive load on a continuously keyed basis throughout the 2 to 30 MHz frequency range. The power output rating in the single sideband modes shall be based on the standard two-tone audio input.

3.16.1.1.1 Voltage standing wave ratio (VSWR) requirements. The equipment shall be capable of operation into a 50 ohm transmission line supplying an antenna system having a VSWR up to 3:1 without damage to the transmitter. The transmitter minimum forward power may be reduced with respect to the minimum levels specified in 3.16.1.1 by two dB at a VSWR of 1.5:1, four dB at 2:1, and six dB at 3:1. The equipment shall not be damaged by operation into a transmission line with a VSWR greater than 3:1 including the conditions of open and short circuit.

3.16.1.2 APC power output leveling. The equipment shall be equipped with an average power control (APC) system which operates to maintain the output power level within +1.5 dB of the minimum outputs of 3.16.1.1 over the frequency range from 2.0 to 30.0 MHz.

3.16.1.3 Modes of operation. Modes of transmitter operation shall be as specified in 3.7.

3.16.1.4 Intermodulation distortion. All intermodulation products of the equipment, when operated as a complete system, shall be reduced at least 30 dB below either tone of a two-tone envelope at a minimum of 100 watts PEP, from 2.0 to 30.0 MHz, produced by using the standard two-tone input (see 6.6.17).

3.16.1.4.1 APC modulation-following distortion. Intermodulation products shall be reduced at least 30 dB below either tone of a two-tone envelope at a minimum of 100 watts PEP with frequency separation of audio input tones reduced to 110 Hz.

3.16.1.4.2 Vibration test distortion. When tested in accordance with 3.15.1.24.2, any vibration related amplitude modulation shall be less than three percent of a single tone RF output of 25 watts. There shall be no breakup of the RF output signal.

3.16.1.5 Harmonic suppression. The suppression of harmonic outputs shall be referenced to the output produced by a single tone in the CW mode (50 watts nominal). Second harmonic output, when measured at 2.0000, 4.0000, and 8.0000 MHz shall be reduced at least 35 dB below the reference. Second harmonic output at all other test frequencies shall be at least 40 dB below the reference. All other harmonic outputs shall be at least 45 dB below the reference.

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3.16.1.6 Carrier suppression. In the USB or LSB mode of operation with the internal carrier insertion switch in the full carrier suppression position, the carrier shall be suppressed a minimum of 50 dB below nominal 25 watts single tone RF output. With the carrier insertion switch in the 0 dB (full carrier), -10 dB, or -20 dB position, the carrier shall be within three dB of the selected level relative to a 25 watt single tone output in USB, LSB, or ISB modes.

3.16.1.7 Opposite sideband suppression. The level of the opposite sideband frequencies (measured at a frequency one kHz or greater from the suppressed carrier) shall be suppressed at least 60 dB below a 1000 Hz single tone signal at 25 watts.

3.16.1.8 Spurious RF emissions. Spurious RF emissions, when measured across a standard 50 ohm resistive load at the transmitter RF output terminals, shall meet the following requirements.

3.16.1.8.1 Out-of-band emissions. All out-of-band spurious emissions, with the exception of harmonics of the primary RF band, specified in 3.16.1.5, shall be reduced at least 50 dB below a single tone output of 25 watts. Spurious emissions shall be reduced at least 50 dB below 25 watts for the condition of a keyed transmitter with no audio modulation (audio lines terminated in 600 ohms).

3.16.1.8.2 In-band products. Spurious emissions which fall within the intelligence band shall be reduced at least 40 dB below a single tone output of 25 watts. This requirement is for either a modulated, or keyed but unmodulated transmitter, and includes frequencies which are produced by modulation of the carrier by harmonics of the tone.

3.16.1.9 Noise. The transmitting system broadband noise output into a 50 ohm resistive termination shall be measured in the USB and LSB modes, and shall not exceed the following levels for the three kHz information band:

a. With the system in a key-down condition, audio inputs terminated in a 600 ohm resistance and with no signal input, the noise output shall be attenuated at least 45 dB below 25 watts average.
b. With the system in a key-up condition, audio inputs terminated in a 600 ohm resistance and with no signal input, the noise output shall not exceed 115 decibels referred to one milliwatt (dBm).

3.16.1.10 Power supply hum distortion. All hum distortion signals shall be suppressed a minimum of 45 dB with respect to an RF output of 25 watts produced by a standard single tone audio input at 1000 Hz.

3.16.1.11 Average power control (APC). An APC circuit shall be provided to maintain the average power output within safe limits when operated as a system, and to maintain the level of the reinserted carrier in an A3h (compatible AM) mode. Time constants of the system APC circuits shall be selected to minimize signal distortion (see 3.16.1.4.1).

3.16.1.12 Power control (PPC). A PPC circuit shall be provided to limit the power output peaks to a level sufficient to safeguard the equipment, including the Antenna Coupler CU-937( )/UR, from damage.

3.16.1.13 CW keying. Keying rates up to 32 baud shall be possible. The keyed wave shall be free from transients and splatter.

3.16.1.14 Keying transient. There shall be no keying transient (signal burst) when the equipment is keyed and unkeyed.

3.16.1.15 Regeneration. There shall be no evidence of unintended oscillations or instability under any condition of equipment operation, including primary line voltage 10 percent higher than the nominal primary line voltage.

3.16.1.16 Oscillator radiation. Oscillator radiation shall conform to the class IB requirements of MIL-STD-461.

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3.16.1.17 Electromagnetic compatibility (EMC). The equipment shall meet the requirements of MIL-STD-461 for Class 1B, C-E equipment. Tests CE01 and CE03 shall be performed at both 60 and 400 Hz power line frequencies.

3.16.1.18 Accelerated life. The equipment shall conform to the requirements of MIL-E-16400 for accelerated life.

3.16.2 Detail requirements for receiver-transmitter unit.

3.16.2.1 General.

3.16.2.1.1 Accuracy of main tuning indicator calibration. The accuracy of frequency indications shall fall within the limits specified herein. The specified accuracies shall apply to all tuning indicators acting in combination, to indicate the tuning of the equipment.

3.16.2.1.1.1 Incremental tuning. Incremental tuning of 100 Hz shall be provided with an accuracy equivalent to that of the internal frequency standard. It shall be possible to tune the equipment to any frequency within the tuning range in less than 20 seconds.

3.16.2.1.1.2 Frequency vernier. A control shall be provided to permit switching into a vernier tuning mode and to provide continuous vernier tuning between one kHz increments in the receive mode. Frequency stability shall be such that for a fixed RF input frequency in SSB modes, autio output frequency will not vary more than 50 Hz in 10 minutes of vernier operation, after a stabilization period of one hour. A flashing amber warning light shall indicate that the vernier control is in the ON position. The vernier control shall be uncalibrated.

3.16.2.1.1.2.1 Vernier override. An override circuit shall be provided to return automatically the transmit mode frequency to a master oscillator locked position when going from receive to transmit mode, even though the tuning in the receive mode is in the vernier condition.

3.16.2.1.2 Frequency and gain stability. The equipment shall include provisions for operation from either an internal or external frequency standard of five MHz. All frequency stability requirements apply to operation using the internal standard.

3.16.2.1.2.1 The following requirements for frequency stability and gain variation shall apply under conditions of A1 or A3h operation as specified. The frequency stability figures shall apply to the variation in a 1000 Hz AF beat note with a constant input frequency in receive or the carrier frequency in transmit. All figures shall refer to total variations, either instantaneous, transient, or gradual. The requirements specified herein shall apply, as pertinent, at any frequency within the frequency range of the equipment.

3.16.2.1.2.1.1 Sixty minute warm-up period. Measurements shall be made in the A1 mode at constant ambient temperature, constant relative humidity and constant line input voltage. Two minutes after turn on time, the equipment shall be within 0.1 percent of the frequency it will have reached at the end of 60 minutes. Twelve minutes after turn on time, the equipment shall be within 0.001 percent of the frequency reached 60 minutes after turn on. Sixty minutes after turn on the equipment shall be within 1 part in 10^7 of the desired frequency. This applies to A1 signals in transmit, or reception of a higher stability frequency signal in a single sideband mode.

3.16.2.1.2.1.2 Variation in primary power supply voltage and frequency. Variation of an AN/URC-35( ) system primary power supply line voltage over a range of 115 volts ±10 percent, and of power line frequency over a range of 48 to 420 Hz, in any combination, shall not cause a change in resonant frequency of the RT-618( )/URC by more than one part in 10^8 (total), or a change in RT-618( )/URC transmit or receive gain in excess of two dB. Frequency requirements apply to transmit A1 signals or reception of a higher stability frequency signal in a sideband mode. Gain requirements apply to the receiver with AGC disabled, and to the transmitter with fixed APC and PPC voltages. For the RT-618( )/URC transmit gain variation test, the system AM-3007( )/URT shall be driven by a constant, externally supplied, unmodulated signal source which causes the system to produce an RF output of 50 watts for 115 VAC, 60 Hz system prime power conditions.

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3.16.2.1.2.1.3 Variation in relative humidity. Exposure of the equipment to MIL-E-16400 humidity conditions shall not cause loss of synthesizer frequency lock or excessive changes in frequency or gain. During the checks performed at the end of the second and fifth cycle of the humidity test, the receiver-transmitter shall be energized for a maximum period of 30 minutes to verify proper operation. At the end of the 30 minute period, A1 transmit frequencies shall be within two parts in 10^7 of the nominal, and maximum permissible gain changes (receiver AGC off, transmitter APC and PPC voltages fixed) shall not produce output variations of more than six dB (total) from the levels recorded as references at the end of the initial dry-out period. During the eight hour operating humidity cycle, transmit A1 frequencies shall be within one part in 10^7 of the nominal after the first hour's operation, and output levels (receiver AGC off; transmitter APC and PPC levels fixed) shall not vary more than six dB total from the reference values for any mode of operation.

3.16.2.1.2.1.4 Variation in ambient temperature.

3.16.2.1.2.1.4.1 Nonoperating condition. The equipment shall be capable of normal operation after exposure to the temperature cycle specified in MIL-E-16400 for temperature tests to be applied to the equipment when in a nonoperating condition.

3.16.2.1.2.1.4.2 Operating condition. The equipment when subjected to the temperature tests for Class 2 (modified) equipment specified in MIL-E-16400 (when in the operating condition) shall not exhibit frequency variations in excess of one part in 10^7. The maximum permissible change in gain for any mode of operation shall be 12 dB. The specified limits for variation of frequency and gain shall be the total deviations of these parameters which occur during the time period required by the temperature cycling test procedure specified in MIL-E-16400. The AGC shall be off during gain variation measurements.

3.16.2.1.2.1.5 Control variations. Frequency variations resulting from a variation of any nonfrequency determining control throughout the full range of operation shall not exceed one part in 10^8. This requirement shall apply to those controls which operate in nonfrequency determining circuits, and to controls which affect the RF or AF gain or level. The specified limit for frequency variation shall be the absolute maximum deviation of this parameter which occurs during the manipulation of any control throughout its full range of operation.

3.16.2.1.2.1.6 Short and long term stability. The equipment shall be capable of operation for a period of 24 hours without operator attention. Without operator attention shall be interpreted as meaning that reliable transmission or reception of a desired signal shall be assured throughout a 24-hour period without the necessity for readjustment of any tuning control. This requirement shall be applicable throughout the complete frequency range of the equipment.

3.16.2.1.2.1.6.1 During the 24-hour period of operation specified in 3.16.2.1.2.1.6, the equipment shall not exhibit sudden, sporadic shifts in the frequency generated by any of the oscillators, or sudden changes in gain that may be the result of easing of tensions in tuning drive mechanisms or from similar effects.

3.16.2.1.2.1.6.2 Frequency drift rate. After an initial stabilization period, not to exceed 48 hours, following turn-on, the overall frequency drift rate shall not exceed one part in 10^8 (total) per day, when adjusted for the reception or transmission of A1 signals at any frequency throughout the frequency range of the equipment. This rate shall be achieved after an initial stabilization of the equipment after turn-on, for a period which shall not exceed 48 hours. The long-term frequency drift rate, averaged over a period of 60 days, shall not exceed 2.5 parts in 10^9 (total) per day. During measurement of frequency drift rate, changes in power supply line voltage and frequency shall be limited to ±1 percent, and temperature variations shall be limited to 10°C (total). Under the same conditions, receiver gain (AGC off) shall not change by more than three dB (total) during a 24 hour period after initial stabilization.

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3.16.2.1.3 Frequency standard input, calibration and adjustment. The equipment shall contain provisions that permit the comparison of an external higher stability frequency standard input of five MHz with the internal frequency standard of the equipment. The comparator shall function with external signal levels between 450 millivolts and 10.0 volts, open circuit, in series with 50 ohms.

3.16.2.1.3.1 Frequency standard calibration. The internal frequency standard shall contain a means of indicating calibration accuracy within one part in 10^8 when compared to an external frequency standard. A dial shall be provided to indicate positioning of the frequency control circuit element within its adjustment range.

3.16.2.1.3.1.1 Frequency standard adjustment range. The internal five MHz frequency standard shall be capable of adjustment over a minimum frequency range of five MHz ±20 Hz. The adjustment shall be accessible from the top surface of the frequency standard assembly without removing the assembly cover. Resolution of the adjustment shall be adequate to permit setting to within one part in 10^9. Piston type adjustments shall be protected from damage at the ends of their adjustment ranges.

3.16.2.1.3.2 Operation using an external frequency standard. The equipment shall be capable of operation from an external frequency standard of five MHz with any input level between 450 millivolts and 10.0 volts, open circuit in series with 50 ohms. When the equipment is operated from an external frequency standard and the level of the externally supplied five MHz signal falls below 450 millivolts, open circuit, in series with 50 ohms, the internally contained five MHz crystal oscillator shall be automatically selected as the reference (see 3.16.2.1.3.4). When the externally supplied five MHz signal is restored to a level greater than 600 millivolts, the external five MHz signal shall again be automatically chosen as the reference. Input impedance at the external five MHz input terminals shall be 50 ohms nominal, within a VSWR of 1.5:1 or less.

3.16.2.1.3.3 Reference output signal. The equipment shall provide a five MHz reference output signal of one volt, open circuit (minimum), in series with 50 ohms, for external use.

3.16.2.1.3.4 Frequency standard switch functions. The internal frequency standard assembly shall have a function switch located on the top of the assembly, accessible without removal of the assembly dust cover. This switch shall have three positions with the following functions (clockwise):

a. EXT (OVEN STBY) (External Oven Standby) - provides operation from an external frequency distribution system (the normal method of operation in installations having such systems): the internal frequency standard oscillator and oven are kept on for immediate availability with full accuracy.
b. EXT NORM (External Normal) - provides operation from an external frequency distribution system, but the internal frequency standard oscillator and oven are turned off.
c. INT/COMP (Internal/Compart) - used when external frequency distribution system is not available or not functioning, or for testing.

3.16.2.1.4 Audio level meter. The equipment shall be provided with a front panel audio level meter that will provide an indication of audio level in the receive or transmit condition. The audio level meter shall be capable of indicating, within ±1 dBm, audio levels from -15 dBm to +20 dBm, over a range of 200 to 5000 Hz. The meter shall be in accordance with MIL-M-10304. Multiple scales may be used to cover the output level range.

3.16.2.1.5 Tune cycle. The voltage from the regulated +20 VDC power supply shall drop to zero VDC and remain there from the start of resetting the one and 10 MHz frequency controls to completion of tuning of the RF amplifier turret and the MHz synthesizer drive motor when the one and 10 MHz frequency controls are tuned at a rate no slower than two steps per second.

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3.16.2.2 Receive mode.

3.16.2.2.1 Sensitivity and gain. *The following sensitivity and gain requirements of* 3.16.2.2.1.1 through 3.16.2.2.1.3 shall apply to the receiver.

3.16.2.2.1.1 Sensitivity. The sensitivity of the equipment, as defined in 6.6.7, shall satisfy the requirements shown in TABLE II for any frequency within the complete frequency range of the equipment. *Specified sensitivities shall be achieved in all modes with the manual RF* gain control in the fully clockwise (maximum gain) position.

TABLE II. Sensitivity - receive mode.

Mode	CW	AM	USB, LSB
IF bandwidth (kHz)	0.400	6.0	3.0
Sensitivity (microvolts)	0.45	6.0	1.2

3.16.2.2.1.2 Gain. The receiver gain shall be sufficient to produce standard audio output (see 6.6.3) for the input voltages of TABLE II, for any frequency within the complete frequency range of the equipment.

3.16.2.2.1.3 Gain variation. *The variation in overall gain for any mode of reception shall* not exceed 10 dB for the complete working frequency range of each megahertz frequency band. Gain variation over the complete equipment tuning range (two to 30 MHz) shall not exceed 15 dB total.

3.16.2.2.2 Selectivity.

3.16.2.2.2.1 Selectivity (antenna input to detector input). The selectivity of the receiver, excluding the audio frequency response characteristics, shall depend primarily upon the characteristics of the filter defining the final IF passband. When measured in accordance with 4.9.2.2.2, selectivity shall be as specified in 3.16.2.2.2.1.1 through 3.16.2.2.2.1.3.

3.16.2.2.2.1.1 USB, LSB.

a. USB passband (minus three dB): carrier plus 300 Hz to carrier plus 3500 Hz.
b. LSB passband (minus three dB): carrier minus 300 Hz to carrier minus 3500 Hz
c. Ripple: Three dB peak to valley, maximum
d. Attenuation at carrier frequency: 15 dB minimum
e. Opposite sideband attenuation one kHz into opposite sideband: 60 dB minimum
f. Attenuation at carrier plus 5100 Hz (USB): 60 dB minimum
g. Attenuation at carrier minus 5100 Hz (LSB): 60 dB minimum

3.16.2.2.2.1.2 AM.

a. Passband (minus three dB): carrier minus 3500 Hz to carrier plus 3500 Hz
b. Ripple (peak to valley): three dB maximum
c. Attenuation at carrier plus 9.5 kHz, carrier minus 9.5 kHz: 60 dB minimum

3.16.2.2.2.1.3 CW.

a. Passband (minus three dB): carrier minus 200 Hz to carrier plus 200 Hz
b. Ripple (peak to valley): three dB maximum
c. Attenuation at carrier plus 500 Hz, carrier minus 500 Hz: 60 dB minimum

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3.16.2.2.2.2 Overall passband (antenna input to audio output). When measured in accordance with 4.9.2.2.2.1, the overall receiver passband characteristics shall be as specified in 3.16.2.2.2.2.1 through 3.16.2.2.2.2.3.

3.16.2.2.2.2.1 USB, LSB.

a. USB passband: carrier plus 300 Hz to carrier plus 3500 Hz
b. LSB passband: carrier minus 300 Hz to carrier minus 3500 Hz
c. Ripple (peak to valley):

USB (carrier + 300 Hz to carrier + 500 Hz: five dB)
(carrier + 500 Hz to carrier + 3500 Hz: three dB)

LSB (carrier + 300 Hz to carrier - 500 Hz: five dB)
(carrier - 500 Hz to carrier - 3500 Hz: three dB)

3.16.2.2.2.2.2 AM

a. AM passband (minus three dB): carrier minus 3500 Hz to carrier plus 3500 Hz
b. Ripple (peak to valley): three dB maximum

3.16.2.2.2.2.3 CW

a. Passband (minus three dB): carrier minus 200 Hz to carrier plus 200 Hz
b. Ripple (peak to valley): three dB maximum

3.16.2.2.2.3 Overload selectivity (desensitization). Overload selectivity shall be such that when the receiver is operated in SSB modes, and with AGC on (slow), an unmodulated interfering signal of 0.4 volt open circuit or less, removed from resonance by five percent of the resonant frequency or more, shall not cause a decrease of more than three dB in the signal-plus-noise to noise ratio (S+N/N) due to an inband signal at the level of standard sensitivity.

3.16.2.2.2.4 Gain depression. At the desensitization points as specified in 3.16.2.2.2.3, gain depression shall not reduce the desired output by more than eight dB.

3.16.2.2.3 Spurious responses. The receiver shall comply with the requirements of 3.16.2.2.3.1 through 3.16.2.2.3.6. The definitions of 6.6 are applicable.

3.16.2.2.3.1 Internally generated spurious outputs. With no RF input signal, the antenna input terminated with a shielded 50 ohm resistor, and the receiver adjusted for the reception of SSB signals, internally generated receiver outputs shall conform to the requirements of 3.16.2.2.3.1.1 and 3.16.2.2.3.1.2.

3.16.2.2.3.1.1 Receiver operated on its internal frequency standard. No more than three spurious signals shall be generated with output S+N/N ratios from 3 to 20 dB. All other spurious signals generated within the reciever shall cause a S+N/N ratio of no greater than three dB. The noise reference shall be that in an adjacent 3.2 kHz spur-free channel. The signal component of the S+N/N ratio shall be that of the spurious signal.

3.16.2.2.3.1.2 Receiver operated on an externally supplied 5 MHz reference. When the receiver is supplied an external five MHz reference at an available power of +21 dBmW (five volts open circuit in series with 50 ohms), having all non-harmonically related components at least 90 dB, and all harmonically related components at least 60 dB below the reference, internally generated spurious outputs shall conform to the requirements of 3.16.2.2.3.1.1, except that the output S+N/N ratio at a receiver operating frequency of five MHz (4.9990 MHz in USB Mode) shall not exceed 45 dB.

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3.16.2.2.3.2 Image rejection. Rejection of the first mixer image shall not be less than 100 dB for any receiver resonant frequency below 20 MHz. First mixer image rejection shall not be less than 90 dB for receiver resonant frequencies of 20 MHz and above. Rejection of second and third mixer images shall not be less than 100 dB.

3.16.2.2.3.3 IF rejection. Rejection of the first IF shall not be less than 100 dB for any receiver resonant frequency below 22 MHz. First IF rejection shall not be less than 80 dB for resonant frequencies of 22 MHz and above. For any operating frequency which differs from the corresponding second IF by more than five percent, second IF rejection shall not be less than 100 dB. Third IF rejection shall not be less than 100 dB for all operating frequencies.

3.16.2.2.3.4 Other single-signal spurious responses. Single-signal spurious responses, other than those already considered, shall comply with the requirements of 3.16.2.2.3.4.1 and 3.16.2.2.3.4.2.

3.16.2.2.3.4.1 Resonant frequency 20 MHz and below. Spurious responses which occur at a signal frequency removed from resonance by 2.5 percent or more of the resonant frequency shall be attenuated by a minimum of 100 dB. Spurious responses which occur at a signal frequency removed from resonance by less than 2.5 percent of the resonant frequency shall be attenuated by a minimum of 60 dB.

3.16.2.2.3.4.2 Resonant frequency above 20 MHz. Spurious responses which occur at a signal frequency removed from resonance by 2.5 percent or more of the resonant frequency shall be attenuated by a minimum of 80 dB. Spurious responses which occur at a signal frequency removed from resonance by less than 2.5 percent of the resonant frequency shall be attenuated by a minimum of 60 dB.

3.16.2.2.3.5 Front-end intermodulation. Signals at two frequencies, f_1 and f_2, shall be applied simultaneously to the antenna input from a 50 ohm linear combiner. Sufficient filtering or isolation shall be used between each generator and the combiner to ensure adequate freedom from intermodulation products in the test signal. Signal pairs shall be $(f_o + \Delta f, f_o + 2\Delta f)$, $f_o - \Delta f, f_o - 2\Delta f)$, $(f_o + 2\Delta f, f_o + 3\Delta f)$ and $(f_o - 2\Delta f, f_o - 3\Delta f)$, with Δf equal to five percent f_o. Signal pairs shall be increased in amplitude, maintaining equality of the two signals until the receiver output signal-plus-noise to noise ratio reaches 10 dB or until each input signal amplitude reaches 0.1 volt open circuit. The receiver output signal-plus-noise to noise ratio shall not reach 10 dB for input signal amplitudes (each signal) of 0.1 volt open circuit or less.

3.16.2.2.3.6 Cross modulation. Cross modulation selectivity shall be such that an off-resonant signal, 90 percent modulated, with a carrier amplitude of 5.0 volts open circuit, or less, and spaced 15 percent or more from the resonant frequency, shall not produce an output greater than 10 dB below that resulting from the resonant signal. For operating frequencies below 25 MHz, this requirement also applies when the frequency separation is 2.5 percent, or more, and the interfering signal carrier level is 0.1 volt open circuit, or less. For operating frequencies above 25 MHz, the requirement also applies for a frequency separation of 4.0 percent, or more, and an interfering signal carrier level of 0.1 volt, or less. AGC shall be on (slow).

3.16.2.2.4 Audio outputs. The audio outputs from the audio amplifier shall conform to the following requirements. The amplifier shall feed line output terminals at the rear of the equipment and its own head telephone jack or local handset connector on the front panel. The line output shall terminate in an appropriately identified receptacle. Transformer coupling shall be provided between the AF amplifier and its associated head telephone jack, local handset connector and AF line output connector. The secondary windings feeding power to the output connector and to the head telephone jack or local handset connector shall be electrostatically shielded from their associated primary windings. Secondary windings which feed AF power to the rear AF line output connector shall be electromagnetically balanced with respect to ground. The center tap shall be brought out to a separate terminal on the pertinent transformer assembly. Grounding of this terminal by soldering methods shall be treated as an installation option, and described in the associated technical manuals.

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3.16.2.2.4.1 Audio output voltage regulation. The audio amplifier shall be so designed that the output voltage shall remain constant within ±30 percent (in volts) of the value obtained with a 600 ohm nonreactive load for a change in the resistive value of this load from 100 to 600 ohms in 100 ohm steps. This condition shall be obtained with any input level from zero input to that which produces full output. The power output transistors shall not be subjected to damage or deterioration at any output load impedance from zero ohms to an open circuit.

3.16.2.2.4.2 Maximum undistorted output. With an RF input of not less than 1000 microvolts, the maximum undistorted output (see 6.6.11) shall not be less than 60 mW dissipated into a 600 ohm resistive load at the remote output terminals of the audio amplifier; or with the line level control set at the maximum output settings, the undistorted output appearing across a 600 ohm resistive load connected to the phone jack shall not be less than 15 mW.

3.16.2.2.4.3 Audio level controls. Two audio level controls shall be provided on the front panel. One control shall adjust remote (line) level, and the other shall adjust only the local (phone) output. Each control shall provide smooth variation input over a minimum range of 40 dB. All controls shall be standard audio (clockwise log) taper.

3.16.2.2.4.4 Output hum and noise.

3.16.2.2.4.4.1 Output signal-to-hum ratio. The output signal-to-hum ratio for AM, USB, or LSB shall not be less than 45 dB for the standard audio output of six mW. This requirement shall apply separately to each output component which is related to the power line frequency, including hum sidebands on the desired audio output. Measurement shall be in accordance with 4.9.2.2.4.4.

3.16.2.2.4.4.2 Output signal to hum-plus-noise ratio. The output signal to hum-plus-noise ratio for AM, USB, and LSB modes, measured in accordance with 4.9.2.2.4.4, shall not be less than 35 dB.

3.16.2.2.5 Manual RF gain control. The manual RF gain control shall have such range as to provide a variation in the overall gain of the RF and IF amplifiers of the receiver of not less than 125 dB. The 125 dB of attenuation shall be obtained in not less than 150 degrees of angular rotation. Excluding the first 10 dB of gain reduction at the clockwise limit of the control, and the last 10 dB of gain reduction at the counterclockwise end of the control, the maximum rate of change of gain shall be 1.2 dB per degree of control rotation. The manual RF gain control shall be inoperative when the receiver AGC switch is in SLOW or FAST.

3.16.2.2.6 Automatic gain control (AGC) characteristics. The AGC threshold shall be adjustable over a minimum range of 1.0 to 5.0 microvolts, open circuit. Production adjustments shall result in the following initial characteristics:

a. The increase in audio output shall not exceed six dB for an RF input change from 1.2 to 5.0 microvolts, open circuit
b. Audio output shall remain constant within a total tolerance of three dB for RF input variations from 5.0 to 100,000 microvolts, open circuit
c. The total harmonic distortion shall not increase above that in 6.6.11.1 at RF input levels up to 100,000 microvolts, open circuit.

3.16.2.2.6.1 The requirements of 3.16.2.2.6 shall apply for the following conditions, singly or in any combination:

a. Any setting of RF tuning within the nominal frequency range of the receiver
b. Any audio gain adjustment producing audio levels at or below maximum undistorted output
c. Any constant level of carrier modulation between 30 and 90 percent
d. Any constant modulation frequency between the limits of 300 and 3500 Hz

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3.16.2.2.6.2 AGC time constants. The AGC time constants shall be as specified in TABLE III.

TABLE III. AGC time constants.

AGC Switch Position	Attack Time (ms)	Hang Time (ms)	Decay Time (ms)
Fast	NLT 10 1/ NMT 15	NLT 12 NMT 15	NMT 5 2/
Slow (SSB Voice, CW, AM)	NMT 20	NLT 800 NMT 1200	NMT 400
OFF	-	-	-

1/ NLT = not less than
2/ NMT = not more than

3.16.2.2.6.2.1 The AGC switch positions shall be completely independent of the mode selector switch positions. All time constants (see TABLE III) shall be switchable to all mode selector switch positions.

3.16.2.2.6.2.2 The vertical location of the AGC switch on the front panel shall be such as to allow for the future addition of a similar control.

3.16.2.2.6.3 AGC linearity. There shall be no evidence of blocking or amplifier saturation at any input level from standard sensitivity up to one volt.

3.16.2.2.6.4 Resonant overload. Audio output power (remote output) at resonant overload (see 6.6.10) shall be a minimum of 60 mW.

3.16.2.2.6.5 Time constant of recovery from surge inputs. The design shall be such that with the receiver operating under standard conditions as for the reception of A1 signals, and with AGC off, a sharp pulse of CW input signal, at resonance, having a minimum amplitude of five volts (to simulate static surge), will not cause a reduction in sensitivity that will persist in excess of 30 milliseconds (ms) after cessation of the input surge.

3.16.2.2.6.5.1 The time constant shall be measured from the time of removal of the input signal pulse until the output noise level has reached 63 percent of the steady-state value. Start of time shall begin at the moment of removal of the input signal pulse.

3.16.2.2.7 Beat frequency oscillator (BFO) control. The BFO control shall operate (when the mode switch is set at CW) so as to permit a variation of the frequency of the beat frequency oscillator of not less than ±3000 Hz.

3.16.2.2.7.1 BFO frequency variation. The variation of frequency versus angular rotation of the BFO shall be substantially linear. The frequency of the BFO shall increase in the clockwise direction and decrease in the counterclockwise direction of the control, using the center of rotation as zero reference.

3.16.2.2.7.2 BFO buffering. The BFO shall be buffered to prevent pulling of the BFO frequency by more than ±10 Hz due to receiver RF input signal amplitudes of 10,000 microvolts or less.

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3.16.2.2.8 In-band intermodulation. When a composite test signal, consisting of two equal sideband voltages which are not harmonically related and are displaced from the carrier by between 300 to 3500 Hz, but from each other by not more than 1000 Hz, is applied to the RF input, any intermodulation product present at the audio output shall not be less than 36 dB below the audio output level of each sideband tone up to a maximum RF input of 0.1 volt per tone. The audio output shall be taken across the remote output into a 600 ohm nonreactive resistance, with an output level of not greater than three volts per tone.

3.16.2.2.8.1 During vibration tests, the receiver shall be operated in USB mode, AGC ON (slow), with a single unmodulated RF input signal, producing an audio output of 1000 Hz. The RF input signal shall be 1000 microvolts, open circuit, and the resulting audio output signal shall be continuously monitored with an oscilloscope and a wave analyzer. Any vibration-related product present at the audio output shall not be less than 30 dB below the reference 1000 Hz output. No signal loss (dropout) exceeding three dB may occur during any vibration test.

3.16.2.2.9 RF input circuit. The RF circuit for the receiver shall be designed for maximum sensitivity to signals from a source with 50 ohm output impedance. The RF input circuits shall provide protection to prevent damage from RF voltage at the input connector of the receiver of 100 volts open circuit (rms). The protective circuit operating threshold shall be between 10 and 20 volts, open circuit in series with 50 ohms, when measured in accordance with the procedure of 4.9.2.2.9. The antenna overload circuit shall also be operated by changes in the megahertz tuning selectors, providing the designed decoupling during rotation of the megahertz selection motors in the receiver. The overload protection circuit shall automatically reset when the overload voltage no longer exists. At least the above degree of protection shall exist with the receiver on or off. Fuses are not acceptable for RF overload protection. Operation of the protection circuit by an out-of-band transmitter shall not produce more than 40 dB desensitization at any receiver operating frequency; measurement shall be in accordance with 4.9.2.2.9. Receiver sensitivity shall not degrade by more than one dB relative to reference measurements made prior to testing in accordance with 4.9.2.2.9.

3.16.2.3 Transmit mode.

3.16.2.3.1 Power output. The RT-618( )/URC shall be capable of transmitting 250 mW PEP in USB or LSB mode, 125 mW carrier in CW mode, and 62.5 mW carrier in AM mode, into a 50 ohm load. These outputs shall be available on a continuously keyed basis, without harm to circuitry, parts and tubes, meeting the technical requirements herein, throughout the frequency range specified in 3.5 and into a 50 ohm resistive load.

3.16.2.3.1.1 Maximum RF output. An output of 250 mW PEP shall be produced before limiting occurs.

3.16.2.3.1.2 Power output leveling. Power output level change, due only to change in the selected output frequency, shall be not greater than 10 dB (total) within any MHz band, or greater than 15 dB (total) over the entire frequency range specified in 3.5. This requirement shall be met with the average power control (APC) voltage fixed at a level which prevents output limiting at the frequency of highest output.

3.16.2.3.1.3 Sideband output equalization. Overall transmit gain shall be adjusted so that when switching from one sideband to the other, the power output shall not change by more than two dB (total).

3.16.2.3.2 Modes of transmit operation. Modes of operation shall be as follows:

3.16.2.3.2.1 CW(A1). CW operation shall be provided on the carrier frequency. A CW hold circuit shall be provided having a hang time of not less than one second and not more than 1.7 seconds so that no relay in the receiver-transmitter will be slaved to the CW key. The keyed CW wave shall be free of transients and splatter to the degree specified under spurious responses. A 1000 ±100 Hz tone keyed at the CW rate shall be applied to the side tone line for local or remote monitoring.

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3.16.2.3.2.2 AM(A3h). Emission in the AM mode of operation shall be comprised of the carrier frequency and the upper sideband portion of the modulator output. An internal control shall be provided to permit adjustment for full carrier level in the AM mode (100 percent modulation). The phone input shall be available on the sidetone line for external monitoring.

3.16.2.3.2.3 Single sideband (A3a/A3j). Emission in either A3a or A3j mode shall consist of either upper or lower sideband (selectable), consistent with the requirements of 3.16.2.3.2.3.1.

3.16.2.3.2.3.1 An internal control shall be provided to permit selection in the single sideband mode of the following nominal levels of carrier insertions:

a. Full carrier (0 dB)
b. -10 dB of carrier suppression
c. -20 dB of carrier suppression
d. Full carrier suppression

Equipment shall be delivered with the control set on full carrier suppression.

3.16.2.3.3 Intermodulation distortion. All intermodulation products shall be reduced at least 35 dB below either tone of a two-tone RF envelope at 250 mW PEP, from 2.0 to 30.0 MHz, produced by using the standard two-tone input (see 6.6.17). With a two-tone output of 100 mW, all intermodulation products shall be reduced at least 40 dB below either tone of the two-tone envelope.

3.16.2.3.3.1 Vibration test distortion. When tested in accordance with 3.15.1.24.2, any vibration related amplitude modulation shall be less than three percent of a single tone RF output of 62.5 mW. There shall be no break-up of the RF output signal.

3.16.2.3.4 *Harmonic suppression.* Second harmonic output shall be reduced at least 40 dB below a single tone output of 62.5 mW. All other harmonic outputs shall be reduced at least 45 dB below 62.5 mW.

3.16.2.3.5 Carrier suppression. In the USB and LSB modes of operation, with the carrier insertion switch in the full carrier suppression position, the carrier shall be suppressed a minimum of 50 dB below a 62.5 mW single tone output. With the carrier reinsertion control in the 0 dB (full carrier), -10 dB, or -20 dB position, the carrier shall be within three dB of the selected level relative to a 62.5 mW single tone output in the USB or LSB modes.

3.16.2.3.6 Opposite sideband suppression. The level of the opposite sideband frequencies (measured at a frequency one kHz or more from the suppressed carrier) shall be suppressed a minimum of 60 dB below the level of a single tone output of 62.5 mW.

3.16.2.3.7 Spurious RF emissions. Spurious RF emissions, when measured across a standard 50 ohm resistive load at the transmitter RF output terminals, shall meet the following requirements.

3.16.2.3.7.1 Out-of-band outputs. All out-of-band spurious emissions, with the exception of harmonics of the primary RF band, shall be reduced at least 50 dB below a single tone output with a level between 10 and 125 mW. Spurious emissions shall be reduced at least 50 dB below 10 mW for single tone output level less than 10 mW, and for the condition of no audio input, audio lines terminated in 600 ohms, and no APC voltage. Adjustment of exciter output level shall be by application of a DC control voltage to the APC input terminal.

3.16.2.3.7.2 In-band products. Spurious emissions which fall within the intelligence band shall be reduced at least 40 dB below a single tone output with the levels specified in 3.16.2.3.7.1. This requirement includes frequencies which are produced by modulation of the carrier by harmonics of the tone.

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3.16.2.3.8 Noise. With the audio input terminated in a 600 ohm resistance, no audio input signal, and no APC or PPC voltages applied, the transmitter broadband noise output into a 50 ohm resistive termination shall be measured, and shall not exceed the following levels for the three kHz information band:

a. -37 dBm (55 dB below 62.5 mW) with the transmitter in the keydown condition
b. -115 dBm with the transmitter in the key-up condition

3.16.2.3.9 Power supply hum distortion. All hum distortion signals (6.6.18) shall be suppressed a minimum of 45 dB with respect to an RF output of 62.5 mW produced by a 1000 Hz standard single tone audio input.

3.16.2.3.10 Audio compression. An audio compression circuit shall be provided to permit operation using signals of +20 dB above standard audio input without a change in the performance described herein. Time constants for the circuit shall be as follows:

a. Attack time (ms): NLT 30, NMT 50
b. Decay time (sec): NLT 1.5, NMT 2.0

3.16.2.3.11 Overall bandpass characteristics. Measured from audio input to RF output, the overall bandpass characteristics shall comply with the following requirements:

a. Response shall be down three dB or less below the peak response from carrier plus 300 Hz to carrier plus 3500 Hz for USB or from carrier minus 300 Hz to carrier minus 3500 Hz for LSB
b. Response shall be down 20 dB or more below the peak response at carrier plus 50 Hz and below and carrier plus 4000 Hz and above for USB, and at carrier minus 50 Hz and above and carrier minus 4000 Hz and below for LSB
c. Maximum peak-to-valley undulations over the 300 to 3500 Hz bandpass shall be three dB

3.16.3 RF amplifier AM-3007( )/URT.

3.16.3.1 Power output. The minimum power delivered to a standard dummy load over the frequency range of 2.0 to 30 MHz shall be at least as specified in TABLE IV, when driven by a source having the maximum power output capability indicated, for the mode of operation being measured.

TABLE IV. Mode of operation.

	SSB (A3j)	Compatible AM (A3h)	CW (A1)
Input (max)	250 mW PEP	62.5 mW carrier	125 mW carrier
Output (min)	100 W PEP	25 W carrier	50 W carrier

3.16.3.1.1 The RF amplifier shall be capable of delivering these power outputs continuously into a 50 ohm load without damage to parts or tubes.

3.16.3.2 Frequency range and tuning. The frequency range shall be from 2.0 to 30.0 MHz. The RF amplifier shall accept coded channel information from the RT-618( )/URC and automatically select the proper operating band and perform under the requirements of this specification.

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3.16.3.3 Modes of operation. Modes of operation shall be as specified in 3.7.

3.16.3.4 Intermodulation distortion. All intermodulation products shall be at least 32 dB below either of two equal tones at RF frequencies corresponding to the standard audio two-tone input. This requirement shall be met at a minimum output of 100 watts PEP throughout the 2.0 to 30.0 MHz frequency range.

3.16.3.5 Harmonic output. The suppression of harmonic outputs shall be referenced to the output, produced by a single tone of 50 watts average power. The second harmonic output shall be reduced at least 35 dB when measured at the five percent points of the two, four, and eight MHz bands. Second harmonic output at all other test frequencies shall be at least 40 dB below the reference. All other harmonic outputs shall be at least 45 dB below the reference.

3.16.3.6 Broadband noise spectral density. Broadband noise output in a key-down condition measured in a three kHz bandwidth over the frequency range of one to 32 MHz shall not exceed -48 dBm.

3.16.3.7 Power supply hum distortion. All hum distortion products shall be suppressed a minimum of 50 dB with respect to a single tone RF output of 25 watts.

3.16.4 Antenna coupler CU-937( )/UR. The CU-937( )/UR shall meet the performance requirements of MIL-A-23547.

3.16.5 Remote radio control set C-9044( )/URC.

3.16.5.1 Input signals. The C-9044( )/URC shall be capable of accepting audio inputs as follows:

	Signal level	Impedance
H-169( )/U	44 mV (nominal)	30 ohms (nominal)
	60 mW (nominal)	600 ohms (nominal)

3.16.5.2 Output signals. The C-9044( )/URC shall be capable of providing the following audio outputs:

RT-618( )/URC	95 millivolts (max)
Loudspeaker	1 watt (max)

3.16.5.3 Frequency response. The audio frequency response of the C-9044( )/URC shall have a variation of no more than one dB from 300 to 3500 Hz.

3.16.5.4 Power requirements. The C-9044( )/URC shall be designed to operate from a DC source of 24 to 30 volts. The keyline voltage shall be a nominal 12 volts DC.

3.16.5.5 Loudspeaker output distortion. The total harmonic distortion in the signal output supplied to the loudspeaker shall not exceed ten percent at a level of one watt.

3.16.6 Interconnection box J-2940( )/URC-35.

3.16.6.1 Programming. The J-2940(( )/URC-35 shall be delivered, wired and jumpered to accommodate the coding for operation with a 10.7 meters (m) (35 foot) whip antenna. Sufficient terminals, wiring and jumpers shall be provided to permit operation with a 7.7 m (25 foot) whip and an AT-1047( )/U (see 3.4.1 (b)).

3.17 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality control system. The contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of; a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied; b) Government product inspection to measure quality of product to be offered for acceptance; c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 General inspection. Inspection shall be as specified in MIL-E-16400 except as otherwise specified herein.

4.2.1 Classification of inspection. The methods of examination and testing of electronic equipment fall within the following classifications:

a. First article inspection (see 4.3)
b. Quality conformance inspection
    1. Production inspection (see 4.4.1)
    2. Production control inspection (see 4.4.2)
    3. Environmental inspection (see 4.4.3)
c. Reliability testing (see 4.10)
d. Preparation for delivery inspection (see 4.11)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examinations and testing necessary to determine compliance with all applicable requirements of this specification, including the tests specified in TABLE V (see 6.3).

4.4 Quality conformance inspection.

4.4.1 Production inspection (group A). Production inspection shall be made on every item offered for delivery. This inspection shall comprise such examination and tests which will prove the workmanship and reveal omissions and errors of the production process such as: functional and performance test at a limited number of points in the required range, tests which detect deviations from design, tests of controls/adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examination and tests shown in group A of TABLE V.

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TABLE V. Examination and tests.

Examination or test	Requirement	Test	First article	Q C INSP A	Q C INSP B	Q C INSP C
Surface examination	3.1.2	MIL-E-16400	X	X		
Weight	3.10	MIL-E-16400	X	X		
Size	3.10	MIL-E-16400	X	X		
Parts & materials	3.1.2	MIL-E-16400	X	X		
Finish	3.1.2	MIL-E-16400	X	X		
Marking	3.15.1.31	MIL-E-16400	X	X		
Workmanship	3.17	MIL-E-16400	X	X		
Weld	3.15.1.27	MIL-E-16400	X		X	
Mechanical design	3.15	MIL-E-16400	X		X	
Drawer orientation	3.15.1.1	4.9.1.25	X		X	
Bonding & grounding	3.15.1.4	MIL-E-16400	X		X	
Corrosion protection	3.15.1.5	MIL-E-16400	X		X	
Painting	3.15.1.5.1	MIL-E-16400	X		X	
Mode selector switch	3.15.1.8.1	4.9.1.26	X		X	
Assembly identification	3.15.1.18	4.9.1.27	X		X	
Soldering	3.15.1.26	MIL-E-16400	X		X	
Conductors identification	3.15.1.30	MIL-E-16400	X		X	
Frequency standard construction	3.15.1.37	4.9.1.28	X		X	
Systems tests						
Power output	3.16.1.1	4.9.1.1	X	X	X	
VSWR	3.16.1.1.1	4.9.1.2	X		X	
Power output leveling	3.16.1.2	4.9.1.3	X	X	X	
Intermodulation distortion	3.16.1.4	4.9.1.4	X	X	X	
Harmonic suppression	3.16.1.5	4.9.1.5	X	X	X	
Carrier suppression	3.16.1.6	4.9.1.6	X	X	X	
Opposite sideband suppression	3.16.1.7	4.9.1.7	X	X	X	
Out of band spurious	3.16.1.8.1	4.9.1.8	X	X	X	
In band spurious	3.16.1.8.2	4.9.1.8	X	X	X	
Noise	3.16.1.9	4.9.1.9	X		X	
Power supply hum distortion	3.16.1.10	4.9.1.10	X	X	X	
APC	3.16.1.11	4.9.1.11	X		X	
PPC	3.16.1.12	4.9.1.12	X		X	
CW keying	3.16.1.13	4.9.1.13	X		X	
Keying transient	3.16.1.14	4.9.1.14	X		X	
Regeneration	3.16.1.15	4.9.1.15	X		X	
Oscillator radiation	3.16.1.16	4.9.1.16	X		X	
Local/remote operation	3.9.1	4.9.1.17	X		X	
Receiver - transmitter tests						
Accuracy of tuning	3.16.2.1.1	4.9.2.1.1	X	X	X	
Tuning time	3.16.2.1.1.1	4.9.2.1.1.1	X		X	
Vernier operation	3.16.2.1.1.2	4.9.2.1.1.2	X		X	
Vernier stability	3.16.2.1.1.2	4.9.2.1.1.2	X		X	
Frequency stability	3.16.2.1.2	4.9.2.1.2	X	X	X	
Gain stability	3.16.2.1.2	4.9.2.1.2	X		X	
Sixty minute warm-up	3.16.2.1.2.1.1	4.9.2.1.2.1	X		X	
Line voltage variation	3.16.2.1.2.1.2	4.9.2.1.2.2	X		X	
Line frequency variation	3.16.2.1.2.1.2	4.9.2.1.2.2	X		X	

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TABLE V. Examination and tests. (Continued)

Examination or test	Requirement	Test	First article	Q C INSP A	B	C
Variation in humidity	3.16.2.1.2.1.3	4.9.2.1.2.2	X			X
Variation in temperature	3.16.2.1.2.1.4	4.9.2.1.2.3	X			X
Control variation	3.16.2.1.2.1.5	4.9.2.1.2.5	X		X	
Frequency drift rate	3.16.2.1.2.1.6.2	4.9.2.1	X		X	
Frequency std calibration	3.16.2.1.3	4.9.2.1.6	X		X	
Adjustment range	3.16.2.1.3.1.1	4.9.2.1.2.6	X		X	
External frequency standard	3.16.2.1.3.2	4.9.2.1.6.1	X		X	
Audio level meter	3.16.2.1.4	4.9.2.1.7	X		X	
Tune cycle	3.16.2.1.5	4.9.2.1.3	X		X	
Receive mode tests						
Sensitivity	3.16.2.2.1.1	4.9.2.2.1	X	X		
Gain	3.16.2.2.1.2	4.9.2.2.1.1	X		X	
Gain variation	3.16.2.2.1.3	4.9.2.2.1.2	X		X	
Selectivity	3.16.2.2.2	4.9.2.2.2	X		X	
Overall passband	3.16.2.2.2.2	4.9.2.2.2.1	X	X		
Desensitization	3.16.2.2.2.3	4.9.2.2.2.2	X		X	
Gain depression	3.16.2.2.2.4	4.9.2.2.2.3	X		X	
Spurious responses	3.16.2.2.3	4.9.2.2.3	X		X	
Internal spurious	3.16.2.2.3.1	4.9.2.2.3	X	X		
Image rejection	3.16.2.2.3.2	4.9.2.2.3	X	X		
IF rejection	3.16.2.2.3.3	4.9.2.2.3	X	X		
Single signal responses	3.16.2.2.3.4	4.9.2.2.3	X		X	
Front end IM	3.16.2.2.3.5	4.9.2.2.3	X		X	
Cross modulation	3.16.2.2.3.6	4.9.2.2.3.1	X		X	
Audio output regulation	3.16.2.2.4.1	4.9.2.2.4.1	X		X	
Maximum undistorted output	3.16.2.2.4.2	4.9.2.2.4.2	X		X	
Audio level controls	3.16.2.2.4.3	4.9.2.2.4.3	X		X	
Signal to hum	3.16.2.2.4.4.1	4.9.2.2.4.5	X		X	
Signal to hum and noise	3.16.2.2.4.4.2	4.9.2.2.4.5	X		X	
Manual RF gain control	3.16.2.2.5	4.9.2.2.5	X		X	
AGC characteristics	3.16.2.2.6	4.9.2.2.6	X	X		
AGC time constants	3.16.2.2.6.2	4.9.2.2.6.1	X		X	
Resonant overload	3.16.2.2.6.4	4.9.2.2.6.2	X		X	
Surge recovery	3.16.2.2.6.5	4.9.2.2.6.3	X		X	
BFO control	3.16.2.2.7	4.9.2.2.7	X	X		
In band IM	3.16.2.2.8	4.9.2.2.9	X		X	
RF input circuit	3.16.2.2.9	4.9.2.2.9	X		X	
Transmit mode tests						
Power output	3.16.2.3.1	4.9.2.3.1	X	X		
Maximum RF output	3.16.2.3.1.1	4.9.2.3.1.1	X		X	
Power output leveling	3.16.2.3.1.2	4.9.2.3.2	X		X	
Sideband equalization	3.16.2.3.1.3	4.9.2.3.1	X		X	
CW hold	3.16.2.3.2.1	4.9.2.3.1	X		X	
Carrier insertion	3.16.2.3.2.3.1	4.9.2.3.3.3	X		X	
Sidetone	3.16.2.3.2.1 & 3.16.2.3.2.2	4.9.2.3.2	X		X	

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TABLE V. Examination and tests. (Continued)

Examination or test	Requirement	Test	First article	Q C INSP A	B	C
Intermodulation distortion	3.16.2.3.3	4.9.2.3.4	X	X		
Harmonic suppression	3.16.2.3.4	4.9.2.3.5	X	X		
Carrier suppression	3.16.2.3.5	4.9.2.3.6	X	X		
Opposite sideband suppression	3.16.2.3.6	4.9.2.3.7	X	X		
Out of band spurious	3.16.2.3.7.1	4.9.2.3.8	X	X		
In band spurious	3.16.2.3.7.2	4.9.2.3.8	X	X		
Noise	3.16.2.3.8	4.9.2.3.9	X		X	
Power supply hum distortion	3.16.2.3.9	4.9.2.3.10	X	X		
Audio compression	3.16.2.3.10	4.9.2.3.11	X		X	
Overall bandpass	3.16.2.3.11	4.9.2.3.12	X		X	
AM-3007 tests						
Power output	3.16.3.1	4.9.3.1	X		X	
Intermodulation	3.16.3.4	4.9.3.2	X		X	
Harmonic output	3.16.3.5	4.9.3.3	X		X	
Broadband noise	3.16.3.6	4.9.3.4	X		X	
Hum distortion	3.16.3.7	4.9.3.5	X		X	
CU-937 tests	3.16.4	4.9.4	X	X		
C-9044( ) tests	3.16.5	4.9.1.17	X	X		
DC operation	3.6	4.9.5	X		X	
Steady state voltage	3.6	4.9.1.22	X		X	
Power, power factor	3.6	4.9.1.24	X		X	
Steady state frequency	3.6.1	4.9.1.22	X		X	
Transient voltage	3.6.1	4.9.1.22	X		X	
Power interruption	3.6.1	4.9.1.23	X		X	
Temperature sequence	3.11	4.9.1.19	X			X
Low temperature	3.11	4.9.1.19	X			X
High temperature	3.11	4.9.1.19	X			X
Humidity	3.11	4.9.1.19	X			X
Salt fog	3.11	MIL-E-16400	X			X
Accelerated life	3.16.1.18	MIL-E-16400	X			
Reliability	3.13	4.10				
Enclosure	3.15.1.3	4.9.1.21	X			X
Shock	3.15.1.24.1	4.9.1.20	X			X
Vibration	3.15.1.24.2	4.9.1.20	X			X
Inclination	3.15.1.24.3	4.9.1.20	X			X
Thermal design	3.15.1.38	MIL-E-16400	X			
EMI	3.16.1.17	4.9.1.18	X			X

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4.4.1.1 Group A tests. The Group A tests shall be performed in accordance with the test methods of 4.9, except as modified by the following:

a. Power output and intermodulation distortion - two, five, 10, 15, 20, 25 and 29.9999 MHz bands
b. Power leveling, signal-to hum ratio and tuning accuracy - two, 15 and 29 MHz bands
c. Harmonic suppression - second and third order, two, 15 and 29 MHz bands
d. Carrier/opposite sideband suppression - 2.1 MHz, USB and LSB modes
e. Spurious emissions - explore three megahertz bands of highest level spurious outputs determined by first article inspection
f. Frequency stability - five MHz

4.4.2 Production control inspection (group B). Production control inspection shall be conducted on a sampling basis as specified. This inspection shall consist of the examination and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examination and tests shown in group B of TABLE V. Production control inspection shall be conducted on inspection lots that have passed the production inspection of 4.4.1. Sampling shall be in accordance with MIL-STD-105, using inspection level S-3 with an acceptable quality level (AQL) of 6.5 percent for each attribute.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from the new lots and shall not lose their identity.

4.4.3 Environmental inspection (group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1. This inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; life tests, simulated service tests; test of the effects of changes of environment (such as extremes of temperature and humidity, effect of salt air); and tests of the effects of shock, vibration and inclination. Environmental inspection shall include the examination and tests shown in group C of TABLE V and shall be performed as approved in accordance with 4.3.

4.4.3.1 Sampling for inspection of equipment. For the first environmental inspection, one equipment from the first month's production, or from the first 50 units produced, whichever is less, shall be selected. For subsequent environmental inspections, one equipment from each successive 100, or one per month if less than 100 are produced monthly, shall be selected. EMI testing shall be conducted only on one of each successive 200 equipments. Tests shall be conducted on lots that have passed the production inspection of 4.4.1.

4.4.3.2 Noncompliance. If a sample unit fails environmental inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of products which were manufactured with the same conditions, materials, processes, and so forth. If the Government inspector considers that the corrective action will not enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.4).

4.4.4 Reinspection of conforming production control and environmental inspection sample units. Unless otherwise specified, sample units which have been subjected to and passed production control or environmental inspections, or both, may be accepted on the contract provided that they are resubjected to and pass production inspection after repair of all defects.

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4.5 Test plan and test procedures. Test plans and test procedures shall be as specified (see 6.2.2).

4.6 Test reports. Test reports shall be as specified (see 6.2.2).

4.7 Test facilities, chambers and apparatus. Test facilities, chambers and apparatus shall be in accordance with MIL-E-16400.

4.8 Performance of test. Performance of tests shall be in accordance with MIL-E-16400 and as specified herein.

4.9 Test methods.

4.9.1 System tests.

4.9.1.1 Power output. Power output measurements shall be made for each mode of transmitter operation on each primary frequency band. Measurements shall be made at the five, 50 and 95 percent points of each primary frequency band.

4.9.1.2 VSWR protection. Tests shall be conducted in the A1 mode with power amplifier RF output coaxial line VSWR of 1.5:1, 2:1, and 3:1 to show compliance with 3.16.1.1.1. VSWR protection shall be measured by terminating the RF amplifier output coaxial line with resistive loads to produce VSWR's of 1.5:1, 2:1, and 3:1. Measure forward and reflected power with a directional watt meter, and monitor the AM-3007 panel meter to detect combinations of VSWR, coaxial line length, and frequency that cause excessive power amplifier plate current.

4.9.1.3 Power leveling. Measurements shall be made to verify compliance with the requirements of 3.16.1.2.

4.9.1.4 Intermodulation distortion. Measurements shall be made in the A3j mode at the nominal end and midband frequencies for each primary frequency band. Intermodulation distortion shall be measured by applying a standard two-tone audio input (see 6.6.17) and monitoring the output developed across a standard dummy load (50 ohm) with a spectrum analyzer. APC shall be adjusted prior to beginning measurements, and no further adjustment shall be made.

4.9.1.4.1 APC modulation following distortion. Measurements shall be made at the RT-618()/URC high and low gain frequencies to show compliance with the requirements of 3.16.1.4.1.

4.9.1.5 Harmonic suppression. Harmonic outputs shall be measured at the end and mid-band frequencies of each primary frequency band.

4.9.1.6 Carrier suppression. Measurement shall be made at the nominal end and midband frequencies of the two, five, 10, 20 and 29 MHz bands.

4.9.1.7 Opposite sideband suppression. Measurements shall be made at the nominal end and midband frequencies of the two, five, 10, 20, and 29 MHz bands.

4.9.1.8 Spurious RF emissions. Each primary frequency band shall be completely explored for evidence of spurious RF emissions which fall outside of the allowable limits of 3.16.1.8.

4.9.1.9 Noise. The transmitter noise output shall be checked on the two, 16 and 29 MHz bands, on both USB and LSB.

4.9.1.10 Output signal-to-hum. Output signal-to-hum ratio shall be measured at 2.1, 17.1 and 29.9 MHz on both USB and LSB. Measurements shall be made for both 60 Hz and 400 Hz power line frequencies.

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4.9.1.11 APC output. The APC output shall be measured at the arithmetic mean frequency of each channel when an unmodulated RF carrier of sufficient amplitude to dissipate 25 watts average power in a 50 ohm standard dummy load has been applied to the RF amplifier input.

4.9.1.12 PPC output. PPC output shall be measured at the arithmetic mean frequency of each channel when an AM, 1000 Hz, 80 percent modulated RF signal, or a standard two-tone signal, is applied to the RF amplifier input.

4.9.1.13 CW keying. CW keying shall be tested by keying the transmitter to develop rated output and photographing the input and output of the RF amplifier as simultaneously displayed on an oscilloscope. All pertinent information regarding gain, frequency, time scale, and so forth shall be recorded on the photograph.

4.9.1.14 Keying transients. The equipment shall be tested to determine compliance with 3.16.1.14.

4.9.1.15 Regeneration. Each primary frequency band shall be carefully explored for evidence of regeneration, under the conditions of 3.16.1.15.

4.9.1.16 Oscillator radiation. Measurements shall be made to verify compliance with 3.16.1.1.

4.9.1.17 Local and remote operation. Tests shall be conducted using the C-1138( )/UR and C-9044( )/URC-35 to verify compliance with 3.9.1.

4.9.1.18 Electromagnetic compatibility test. Compliance with 3.16.1.17 shall be verified by tests performed in accordance with MIL-STD-462.

4.9.1.19 Temperature and humidity. Tests shall be conducted in accordance with MIL-E-16400. Measurements of gain and frequency stability shall be made to determine compliance with the requirements of 3.16.2.1.2. Measurements shall be performed in USB, LSB and CW modes for a minimum of one high band and one low band frequency. Frequency accuracy (synthesizer lock) checks shall be performed during temperature operating tests at the end of the period of operation at 0°C, after four hours of operation with the equipment stabilized at +50°C, and after return to room temperature. During the last hour of the high temperature operating test, air flow produced by forced convection in the test chamber shall be eliminated. During this period, air temperature in the immediate vicinity of the equipment under test shall be controlled to 50°C by radiation-baffled heating and cooling sources as required. This test is intended to simulate the worst case shipboard thermal environment (see 4.9.1.19.1). During humidity tests, measurements shall be made at the end of the second and fifth cycles (see 3.16.2.1.1 and 4.9.2.1.1).

4.9.1.19.1 Chamber temperature control provisions. The sources for maintaining the chamber environment shall be arranged or suitably baffled so that no surface at a temperature other than the desired ambient is visible to the unit under test. The maximum allowable temperature difference between any two points in the chamber shall be 10°C. The chamber control provisions shall be capable of maintaining the temperature indicated by a selected chamber ambient sensor constant within ±1°C of any selected test temperature.

4.9.1.20 Shock, vibration, and inclination. Prior to and after completion of shock, vibration and inclination tests, the following tests shall be performed:

a. Power output and levelling measurements in accordance with 4.9.1 and 4.9.2 and the A1 and A3j modes for the two, 17, and 29 MHz bands only
b. Synthesizer lock checks in accordance with 4.9.2.1.1
c. Intermodulation distortion in accordance with 4.9.1.4. During vibration test, equipment operation and monitoring shall be in accordance with requirements of 3.16.1.4.2 and 3.16.2.2.8.1. Continuous key-down operation is required during all tests, unless otherwise specified herein.

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4.9.1.21 Enclosure. Tests shall be conducted in accordance with MIL-E-16400. Measurements of gain and frequency stability shall be made prior to and after testing. Measurements shall be within the limits established in 3.16.2.1.2.

4.9.1.22 Supply line voltage and frequency variation and transients. Tests shall be performed in accordance with MIL-E-16400. Reference power output, gain and frequency measurements shall be made prior to start of testing. Variation of transmitter frequency and gain with line voltage and frequency shall be measured against the requirements of 3.16.2.1.2 for nominal line voltage of 115 volts AC, and for nominal line frequencies of 60 and 400 Hz. Following performance of the tests specified in MIL-E-16400, no permanent changes shall be produced in power output, frequency and gain.

4.9.1.23 Power interruption. In addition to the tests specified in MIL-E-16400, a test shall be conducted to verify operation with high voltage transients of three and one-half times nominal voltage but of short duration (less than one cycle). No permanent changes shall be produced in power output, frequency or gain as a result of this test.

4.9.1.24 Power and power factor. Tests shall be conducted in accordance with MIL-E-16400.

4.9.1.25 Drawer orientation. Verification of the ability to orient the drawers ±90 degrees shall be performed.

4.9.1.26 Mode selector switch. Mode selector switch functions specified in 3.15.1.8.1 shall be verified.

4.9.1.27 Assembly identification. All applicable assemblies shall be inspected to verify compliance with 3.15.1.18.

4.9.1.28 Frequency standard construction. Frequency standard assemblies shall be inspected to verify compliance with 3.15.1.37.

4.9.1.29 Accelerated life . Measurements of power output and frequency stability shall be made in accordance with 4.9.1.1 and 4.9.2.1.2 in the two, 17, and 29 MHz bands to verify compliance with MIL-E-16400. A five minute transmit and a five minute receive duty cycle shall be used.

4.9.2 Receiver-transmitter tests.

4.9.2.1 General.

4.9.2.1.1 Accuracy of main tuning indicator calibration. Measurement shall be made in the A1 mode at frequencies required to verify proper operation of each frequency selector in each digital position.

4.9.2.1.1.1 Tuning time. Measurement shall be made of the time required to switch from 2.0000 to 29.5559 MHz.

4.9.2.1.1.2 Frequency vernier range. Measurements shall be made to verify compliance with 3.16.2.1.1.2.

4.9.2.1.2 Frequency and gain stability. Frequency or gain stability or both shall be measured in accordance with the following conditions.

4.9.2.1.2.1 Sixty minute warm-up period. The warm-up period shall begin after exposure of the equipment to the ambient temperature for a period of not less than six hours with no power applied, to establish the temperature at the same value as the ambient. Frequency measurements shall be made at intervals of not more than two minutes for the first 12 minutes, with the first measurements made two minutes after power is applied. Measurements shall be made at intervals of not more than five minutes for the next 18 minutes, and at intervals of not more than 10 minutes for the last 30 minutes.

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4.9.2.1.2.2 Line voltage and frequency variation. Meaurements shall be in accordance with the procedure specified in MIL-E-16400. A minimum stabilization period of 30 minutes shall be observed following each change of supply voltage and frequency.

4.9.2.1.2.3 24-hour frequency drift. All equipment power shall be removed for a period of at least six hours prior to stability measurements. The equipment shall be energized for a stabilization period not to exceed 48 hours, and operate during the test period with temperature, humidity, and line voltage controlled as specified in 3.16.2.1.2.1.6.2.

4.9.2.1.2.4 24-hour gain stability. Gain stability shall be measured in A3j modes. Equipment conditioning periods, and temperature, humidity and line voltage control, shall be as specified in 4.9.2.1.2.3.

4.9.2.1.2.5 Control variations. Measurement shall be made to verify compliance with 3.16.2.1.2.1.5.

4.9.2.1.2.6 Frequency standard adjustment range. Measurement shall be made to verify compliance with 3.16.2.1.3.1.1.

4.9.2.1.3 Tune cycle. Measurements shall be made to verify compliance with 3.16.2.1.5.

4.9.2.1.4 Regeneration. Each primary frequency band shall be carefully explored for evidence of regeneration.

4.9.2.1.5 Oscillator radiation. Measurements shall be made at the nominal end frequencies of each primary (MHz) band.

4.9.2.1.6 Frequency standard calibration. Measurements shall be made to determine compliance with the requirements of 3.16.2.1.3 through 3.16.2.1.3.4.

4.9.2.1.6.1 External frequency standard. Operation using an external frequency standard shall be made to verify compliance with 3.16.2.1.3.2.

4.9.2.1.7 Audio level meter. Measurements shall be made to verify compliance with 3.16.2.1.4.

4.9.2.2 Receive mode tests.

4.9.2.2.1 Sensitivity. Measurements shall be made at the five, 50, and 95 percent points in each of the primary frequency bands for the USB and LSB modes, and at the poorest USB and LSB sensitivity frequencies for the remaining modes of operation.

4.9.2.2.1.1 Gain. Measurements shall be made in each megahertz frequency band to ensure the gain is sufficient to produce standard output.

4.9.2.2.1.2 Gain variation. Measurements shall be made using the procedure of 4.9.2.2.1.2.1 to determine the gain variation of each megahertz frequency band. Measurements shall be made with the receiver in USB mode, with AGC off and RF gain maximum.

4.9.2.2.1.2.1 Gain variation measurement procedure. Apply a signal of one microvolt, open circuit, in series with 50 ohms, at a frequency of 2.1010 MHz, through the receiver antenna input terminal. Tune the receiver to 2.1000 MHz in USB mode. Adjust the RF gain control for maximum and turn AGC to OFF. Adjust the audio line level control to produce six mW of audio output. Leaving the gain and level controls and the input signal amplitude fixed, tune the receiver and signal source together to produce a constant output frequency of one kHz. Within each megahertz frequency band, note output for each setting of the 100 kHz selector, with all less significant selectors set at zero. Continue procedure with each selector until maximum output is determined and recorded for each MHz band. Repeat the procedure, determining and recording minimum output in each band.

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4.9.2.2.2 Selectivity (antenna input to detector input). Measurements shall be made in USB, LSB and AM modes with the receiver tuned to 17.5500 MHz. AGC shall be inoperative and RF gain shall be maximum. Measurements shall be made of all parameters specified in 3.16.2.2.1.1 and 3.16.2.2.1.2 using an unmodulated test signal. Ripple in the passband shall be measured between the -3 dB points using a constant amplitude RF input signal which produces an output S+N/N ratio of 20 dB at a peak of the passband ripple. A record shall be made of the amplitude and frequency (relative to the carrier) of each inflection point. Measurements of carrier and opposite sideband rejection and 60 dB attenuation points shall be made by increasing the RF input signal amplitude to obtain the reference IF output produced at the peak of passband ripple.

4.9.2.2.2.1 Overall passband (antenna input to audio output). Measurements shall be made in USB, LSB and AM modes, with the receiver tuned to 17.5500 MHz. AGC shall be inoperative and RF gain shall be maximum. Bandwidth (-3 dB) and passband ripple shall be measured using a constant amplitude RF input signal which produces an output S+N/N ratio of 20 dB at the peak of the passband ripple. The RF input signal shall be unmodulated for USB and LSB modes, and shall be 90 percent amplitude modulated at 1000 Hz for the AM mode.

4.9.2.2.2.2 Overload selectivity (desensitization). Overload selectivity (desensitization) shall be measured with the receiver operating in the SSB mode, and with AGC on (slow). An unmodulated signal shall be applied at resonance to the input of the receiver from a 50 ohm standard dummy antenna and adjusted to yield standard sensitivity. An unmodulated interfering signal shall be applied from the same standard dummy antenna at a frequency differing from the original signal by the IF bandwidth; the signal level shall be increased to the point where the signal-plus-noise to noise ratio of the original signal is decreased three dB. The interfering signal level shall be increased in steps, up to a maximum which is 120 dB above the specified sensitivity level while the frequency separation is increased to maintain the three dB decrease in signal-plus -noise to noise ratio.

4.9.2.2.2.3 Gain depression. At each point on the overload selectivity curve, the amount of gain depression for the desired signal shall be measured at the three dB desensitization point by means of a wave analyzer tuned to the desired audio output frequency.

4.9.2.2.3 Spurious responses. Each primary frequency band shall be completely explored for evidence of the following types of spurious responses which fall outside of the allowable limits as indicated in 3.16.2.2.3.

a. Internally generated spurious outputs (AGC off)
b. Image rejection
c. IF rejection: measurements shall be made at the nominal end frequencies of each primary band
d. Other single-signal spurious responses
e. Front-end intermodulation (2.4, 17.1, and 26.1 MHz)

4.9.2.2.3.1 Cross modulation selectivity. Cross modulation selectivity shall be measured with AGC on (slow), by applying a 1000 microvolt signal that is modulated 90 percent at 1000 Hz at the resonant frequency to which the receiver is tuned, and adjusting the gain of the set to produce 60 mW of AF output (across the line output connector). The modulation of the desired signal shall then be removed and an interfering off-resonant signal, modulated 90 percent at 1000 Hz, shall be applied and adjusted in frequency and amplitude to produce a cross modulation AF output level of six mW without any change in the control settings of the receiver. This interfering signal shall then be readjusted, in both frequency and amplitude, on both sides of the resonant frequency to which the receiver is tuned, so as to produce six mW of interfering output as the result of cross modulation at the various frequencies off resonance. The interfering signal input level shall range from a value which begins to produce the cross modulation phenomenon at the closest possible spacing to the desired signal frequency, up to interfering signal levels of at least five volts. The cross modulation selectivity shall not be degraded for desired signal inputs up to at least 60 dB above standard sensitivity input. The cross modulation performance shall be measured in the A3 mode. Cross modulation selectivity shall be measured at four points; that is, f_o ±15 percent, and either f_o ±2.5 percent, or f_o ±4 percent, as appropriate. Measurement shall be made for operating frequencies of 2.1, 24.9, and 29.9 MHz.

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4.9.2.2.4 Audio output measurements.

4.9.2.2.4.1 Audio output voltage regulation. Measurements of audio output variation with change of load impedance and with variation of input signal level shall be made at the line output of the audio amplifier. Tests shall be performed to ensure that no damage to output transistors occurs when operated into short and open circuit loads under zero drive and maximum drive conditions.

4.9.2.2.4.2 Maximum undistorted output. Measurements shall be made in the AM, USB, and LSB modes at a nominal RF input frequency of 2.1 MHz. The AGC shall be operative, and measurements shall be made for all positions of the AGC time constant selector switch. Measurements shall be made on both rear mounted line output receptacles and panel mounted head telephone jacks. AF output data shall be obtained for applied RF signal inputs from 1000 microvolts up to one volt, with the input level recorded in decade steps of one decade each.

4.9.2.2.4.3 Range and linearity of audio level controls. Measurements shall be made of line output and head telephone control characteristics in the USB mode. AGC shall be active in the SLOW position of the AGC switch. Audio output as a function of angular rotation of the audio gain controls in increments of 10 degrees or less shall be recorded starting from the fully CCW position. The measurements shall be performed at a nominal operating frequency of 2.1000 MHz.

4.9.2.2.4.4 Output hum and noise. Measurements shall be made for AM, USB, and LSB modes at RF input frequencies of 2.1, 17.1, and 29.9 MHz. The input signal level shall be 1000 microvolts, supplied from a standard 50 ohm dummy antenna. For single sideband modes, the signal shall be unmodulated; for AM, the signal shall be modulated 90 percent at 1000 Hz. Measurements shall be made for power line frequencies of 60 Hz and 400 Hz, as follows:

a. Set receiver RF gain to maximum and adjust line level control to produce standard output of six mW. With this level as a reference, measure relative amplitude of power line related components in the output, using a wave analyzer
b. Set receiver line level control to maximum gain, and adjust the manual RF gain control to produce standard output of six mW. With this level as a reference, remove the RF input signal and measure the relative amplitude of the remaining hum and noise components with a broadband voltmeter

4.9.2.2.5 Manual RF gain control. Range and linearity measurements shall be made with the receiver conditioned for USB (AGC inoperative) for an applied nominal RF input frequency of 2.1 MHz. Record gain reduction in dB as a function of angular rotation in increments not greater than 10 degrees.

4.9.2.2.6 AGC characteristics. Receiver AGC amplitude characteristics and time constants shall be measured for USB, LSB, and AM operation at the receiver low gain frequency. AGC linearity shall be verified. Measurements shall be made of maximum change in audio output level for RF inputs between that which produces standard output S+N/N in the SSB modes and one volt, open circuit. No evidence of blocking or saturation of output signal shall occur.

4.9.2.2.6.1 AGC time constants. Separate determinations of each of the AGC time constants shall be made by switching the RF input amplitude between the level which produces standard output response and various large signal levels from 100 microvolts to 0.1 volt, open circuit, and observing the audio output.

4.9.2.2.6.2 Resonant overload. Measurement shall be performed in the USB, LSB, and AM (90 percent modulation) modes at the receiver high gain frequency. The receiver AGC shall be disabled, and AF and RF level controls shall be set for maximum gain.

4.9.2.2.6.3 Time of recovery from surge inputs. Measurement shall be made in USB and LSB modes at the receiver high gain frequency.

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4.9.2.2.7 BFO control. The output frequency range of the BFO control shall be measured at an RF input frequency of 2.1 MHz.

4.9.2.2.8 In-band intermodulation. Measurements shall be made at receiver high and low gain frequencies, in USB and LSB modes, and for FAST and SLOW AGC modes. RF signal frequencies shall be such that audio output tone frequencies are 935 and 1045 Hz. Measurements shall be performed with RF signal amplitudes of 0.37 dBm and 0.13 dBm per tone, and audio output of three volts, rms, per tone.

4.9.2.2.9 RF input circuit . Threshold of operation of the antenna input protection circuit and insertion loss in the operated condition shall be measured at the mid-frequency in each primary (MHz) band to show compliance with requirements of 3.16.2.2.9. Receiver input circuit protection shall be verified in each primary (MHz) band. Receiver input circuit protection shall be verified by keying drive to a suitable RF power amplifier held in a key-down condition. The power amplifier shall provide not less than 100 volts open circuit in series with 50 ohms, and the protection circuit shall be operated a minimum of 10 times at each test frequency. Circuit operating threshold shall be measured at the center of each MHz band. Desensitization produced by operation of the circuit by an out-of-band transmitter shall be measured at the center of each MHz band, simulating an overload by opening the normal connection to antenna overload board terminal A2A9E6, substituting a ground, and then measuring the change in sensitivity.

4.9.2.3 Transmitter mode tests.

4.9.2.3.1 Power output. Power output measurements shall be made at the low gain frequency of each MHz band in USB, LSB, ISB, CW and AM modes with no APC or PPC voltages applied. Standard two tone audio shall be used for USB and LSB modes. A standard single tone audio input to each of the sidebands shall be used for ISB measurements. The unmodulated carrier shall be measured in AM mode.

4.9.2.3.1.1 Maximum RF output . Maximum RF output shall be measured to verify compliance with 3.16.2.3.1.1.

4.9.2.3.2 Power output leveling. Key the transmitter in the CW mode. Adjust DC bias voltage applied to the APC line to hold the RF output power into a 50 ohm load below voltage limiting level (250 mW). For each MHz band record the setting of the 100 kHz, 10 kHz, 1 kHz and 100 Hz frequency controls that produce the highest and lowest RF power. After the transmitter has reached a stable operating temperature (on for at least 4 hours), measure RF output power at all of the frequencies recorded in the previous step with APC voltage held constant within 0.001 volt DC. Calculate the dB power difference between the high and low gain frequency in each MHz band and the difference between the overall high and low gain frequency.

4.9.2.3.2.1 Sideband output equalization. Equalization of overall transmit gain in each sideband shall be measured with APC adjusted to prevent limiting. A single tone shall be placed in each sideband at the peak of the passband ripple.

4.9.2.3.3 Modes of operation. Operation in all modes shall be verified.

4.9.2.3.3.1 CW hold characteristics. The transmitter shall be tested for proper operation of the CW hold characteristic in accordance with 3.16.2.3.2.1.

4.9.2.3.3.2 Sidetone output. Audio sidetone outputs shall be checked in each mode of operation to ensure proper operation.

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4.9.2.3.3.3 Carrier reinsertion. Carrier reinsertion level shall be measured to verify compliance with 3.16.2.3.5.

4.9.2.3.4 Intermodulation distortion. Intermodulation distortion shall be measured by applying a standard two-tone audio input (see 6.6.17) and monitoring the output developed across a standard dummy load (50 ohm) with a spectrum analyzer. Measurements shall be made in the A3j mode using the standard two-tone audio input of 6.6.17. Measurements shall be made at the five, 50 and 95 percent points of each primary frequency band.

4.9.2.3.5 Harmonic suppression. Harmonic outputs shall be measured at the five, 50 and 95 percent points of each primary frequency band.

4.9.2.3.6 Carrier suppression. Measurements shall be made at 2.1, 17.1 and 29.9 MHz to verify compliance with the requirements of 3.16.2.3.5.

4.9.2.3.7 Opposite sideband suppression. Measurements shall be made at 2.1, 17.1 and 29.9 MHz to verify compliance with the requirements of 3.16.2.3.6.

4.9.2.3.8 Spurious RF emissions. Each primary frequency band shall be completely explored for evidence of spurious RF emissions which fall outside the allowable limits of 3.16.2.3.7. Explore each band with APC voltage adjusted for 10 mW single tone RF output with standard single tone audio input. Set selected frequency for maximum amplitude at spurious signal. For spurious signals which change in amplitude with change of APC voltage or audio input voltage, make additional measurements to determine if level exceeds the limits of 3.16.2.3.7.

4.9.2.3.9 Noise. The transmitter noise output shall be measured on the overall high and low gain frequencies and on two other MHz bands on both USB and LSB to verify compliance with the requirements of 3.16.2.3.8.

4.9.2.3.10 Output signal-to-hum ratio. Output signal-to-hum ratio shall be measured at 2.1, 17.1, and 29.9 MHz, on both USB and LSB. Measurements shall be made for both 60 Hz and 400 Hz power line frequencies.

4.9.2.3.11 Audio characteristics. The audio circuits shall be checked for frequency and gain characteristics. Speech processing shall be checked to ensure that an essentially constant amplitude of modulation, allowing for changes due to speech factors, is maintained for variation (within the limits of microphone input voltages) in speech intensity and such variation shall not cause variations in power output.

4.9.2.3.12 Overall bandpass. Measurements shall be made to verify compliance with 3.16.2.3.11.

4.9.3 AM-3007( )/URT tests.

4.9.3.1 Power output. Measurements shall be made for each mode at the nominal end frequencies and at the arithmetic mean of each frequency channel into which the overall frequency range of the RF amplifier has been divided.

4.9.3.2 Intermodulation distortion. Measurements shall be made at the nominal end frequencies and at the arithmetic mean of each frequency channel into which the overall frequency range of the RF amplifier has been divided.

4.9.3.3 Harmonic suppression. Harmonic outputs shall be measured at the end and mid-frequencies of the RF amplifier frequency bands.

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4.9.3.4 Spectral density. Measurements shall be made to verify compliance with 3.16.3.6.

4.9.3.5 Hum distortion. Measurements shall be made to verify compliance with 3.16.3.7.

4.9.4 CU-937( )/UR tests. Tests of a complete AN/URT-24( ), including the CU-937( )/UR, shall be conducted to verify compliance with MIL-A-23547.

4.9.5 DC operational tests. The contractor shall perform such tests as necessary to demonstrate satisfactory operation of the equipment, including transmitter power output, using an external power input source of 27.5 volts DC.

4.10 Reliability testing. Reliability testing shall be performed in accordance with MIL-STD-781. The reliability demonstration shall be performed after first article approval.

4.10.1 Reliability demonstration (qualification) tests. Reliability demonstration testing shall be performed on at least six production units. Testing shall be performed at Test Level C and proceed to a decision in accordance with Test Plan IV of MIL-STD-781. The AN/URC-35 reliability demonstration shall be continued beyond the accept decision point as required to verify the separate CU-937( )/UR reliability requirement of MIL-A-23547. All reliability demonstration requirements shall be successfully completed prior to delivery of any production equipments under the contract.

4.10.1.1 Failure. Failure shall be any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements specified herein (for definition of failure, (see 6.6.20). Any adjustments or alignments after test initiation shall be considered as a malfunction.

4.10.1.2 Rejected equipment. When equipment has been rejected as a result of the reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer design, corrective action shall be performed on all equipment delivered under the contract.

4.10.1.3 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and reconditioning as required and shall be subjected to the production tests of 4.4.1 to ensure that it conforms to the requirements of this inspection. After reconditioning, reinspection, and retesting, the equipment may be delivered as part of the contract quantity provided it conforms to all terms and conditions of the contract for acceptance.

4.10.1.4 Reliability test procedures. The reliability test shall be conducted using the procedures, schedules, and performance limits specified herein, as a minimum.

4.10.1.4.1 Input line voltage. Input line voltage shall be as specified in Test Level C of MIL-STD-781. The input voltage shall be cycled in accordance with MIL-STD-781.

4.10.1.4.2 Test frequencies. During the reliability test, at least one frequency shall be tested in each band as shown in TABLE VI:

TABLE VI. Test frequency.

F1	2.0015 MHz
F2	5.3340 MHz
F3	10.6655 MHz
F4	29.9990 MHz

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4.10.1.4.3 Test schedule. The test schedule of TABLE VII, including the input line voltage schedule of 4.10.1.4.1, shall be conducted. The test shall consist of three 8-hour schedules per day, the first being manned during which performance measurements shall be made. The second two may be unmanned and pertinent data will be automatically recorded only to the extent that instrumentation permits. Schedules shall be as shown in TABLE VII. The operating duty cycle during all operating periods (except while performance measurements are taken) shall be 15 minutes transmit and 30 minutes receive.

TABLE VII. Test schedule.

Manned schedule	Unmanned schedule
0-2 hours Radio ON	0-3 hours Radio ON
2-3 hours (See 4.10.1.4.4 Performance Measurements)	3-4 hours Radio OFF
3-4 hours Radio OFF	4-7 hours Radio ON
4-6 hours Radio ON	7-8 hours Radio OFF
6-7 hours Radio measurements	
7-8 hours Radio OFF	

4.10.1.4.3.1 Manned operating schedule. The mode and frequency of operation for each of the operational ON periods of the manned schedule shall occur in the following sequence. During each period, an appropriate RF signal shall be used to simulate actual receiver operation in the particular operating mode under test.

TABLE VIII. Operating schedules.

Day	Frequency	Mode of transmission
1	F1 - First 4 hours of manned schedule F2 - Remainder of the day	CW(A1) 50 W average power keyed at a rate of 32 baud
2	F3 - First 4 hours of manned schedule F4 - Remainder of the day	AM(A3h) 25 watts carrier modulated 30-50% at 1000 Hz
3	F1 - First 4 hours of manned schedule F2 - Remainder of day	SSB(A3a) 100 W PEP, two-tone modulation
4	F3 - First 4 hours of manned schedule F4 - Remainder of day	SSB (A3a) 100 W PEP, two-tone modulation

The same sequence shall be used on the next 4 days, except that the sequence of frequencies shall be reversed, F4 and F3 on the 5th day, F2 and F1 on the sixth day, F4 and F3 on the 7th day and F2 and F1 on the 8th day. The schedule shown in TABLE VIII shall be repeated as necessary.

4.10.1.4.3.2 Unmanned operating schedule. The mode of operation during the unmanned periods shall be SSB(A3a). Voice tape modulation, 100 watts PEP, shall be used during transmitting periods. A corresponding modulated (1000 Hz) RF signal shall be applied to the antenna port to simulate actual receiver operation. The test frequency to be used during each unmanned period shall be the last frequency utilized during the previous manned period (see 4.10.1.4.3.1).

4.10.1.4.4 Minimum performance limit. During each of specified periods of performance measurements, the contractor shall provide necessary tests to assure that equipment performance is not less than the following:

a. CW (A1) mode
   1. Keying rate shall be at least 32 baud
   2. Output shall be at least 50 watts +1.5 dB, -1.0 dB average power into a standard dummy load in key down condition
   3. Keyed wave shall be free from transients and splatter

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b. AM (A3h) mode
   1. No adjustment of the internal carrier reinsert control shall be made during the test providing carrier power level remains within 25 watts +1.5, -1.0 dB into a 50 ohm load. If carrier power variations exceed above limits, sufficient adjustment range of the carrier reinsert control to adjust the carrier power to 25 watts shall be demonstrated.
   2. It shall be possible to apply an audio tone of 1000 Hz to the upper sideband input and adjust its level so that the sideband output equals the carrier (25 watts +1.5, -1.0 dB). Without changing any controls on the receiver-transmitter, the sideband power shall not vary more than 3 dB when the sideband frequency is changed from 300 to 3500 Hz and the input level is maintained constant at the level required for equal sideband and carrier output at 1000 Hz input.
   3. Under the test conditions of step 2, the intermodulation distortion products shall be at least 27 dB below the carrier.

c. SSB (A3a) mode
   1. With a standard two-tone input applied to the local input, the output of the amplifier shall be 100 watts PEP +1.5, -1.0 dB into a standard dummy load
   2. With modulation frequencies of 1000 and 1625 Hz, the intermodulation distortion shall not be less than 27 dB below either tone of a two-tone envelope at 100 watts PEP (+1.5 dB, -1.0 dB) when measured into a standard dummy load
   3. Opposite sideband suppression shall be at least 60 dB when a 50 mV, 1000 Hz tone inserted into the upper sideband local input is adjusted for 25 watts output into a standard dummy load
   4. Carrier suppression shall be adjustable to at least 50 dB below a sideband tone of 1000 Hz having an input level of 50 mV applied to the upper sideband local input

d. Receiver test. The receiver gain and sensitivity shall not decrease below a level requiring more than 1 microvolt of CW input (applied to the antenna input through a standard dummy antenna) in the single sideband mode of operation to produce not less than 6 mW of audio output into a 600 ohm nonreactive load connected across the nominal 600 ohm audio output line at a signal-plus-noise to noise ratio of 10 dB. Receiver overall audio distortion shall not exceed 5 percent of voltage at standard output.

e. Frequency stability. The drift of the reference oscillator (internal frequency standard) shall be not worse than one part in 10^7 per day nor one part in 10^6 per month

f. CU-937( )/UR tests. Power output, VSWR and rough and fine tuning functions shall be verified.

4.10.1.5 Burn-in. The contractor may elect to operate the reliability test samples for a certain time prior to the initiation of reliability demonstration testing, in addition to the burn-in period specified in 3.13.2. Any such burn-in applied to test samples must be applied to all items of equipment submitted for acceptance. Failures occurring during this period may be considered nonrelevant but shall be recorded and reported. Burn-in time for all items of equipment submitted for acceptance shall be at the same levels of environment and for the same integral number of cycles as the burn-in conducted on the reliability demonstration test samples.

4.10.2 Equipment burn-in. All deliverable equipments and spares shall be subjected to the temperature burn-in requirements of 3.13.2. The burn-in shall be performed at a stabilized 50°C ambient temperature with 1.0g ±10 percent vibration applied at least 10 minutes out of each operating hour for a cumulative operating time of at least 24 hours. Failures occurring during the last 24 hours shall be repaired and burn-in resumed for a minimum of 24 hours without further failure.

4.10.2.1 Equipment operation during burn-in. Equipment operation need not be continuous as long as the accumulated operating time conforms to conditions of 4.10.2. The equipment shall be operated in accordance with 3.13.2.

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4.10.2.2 Performance monitoring. Equipment shall be checked for proper performance at least twice every 24 hours and just prior to end of burn-in.

4.10.3 Failure analysis. All failures shall be analyzed and the results of analysis shall be retained by the contractor and be available for Government review as required.

4.11 Preparation for delivery inspection. Inspections shall be conducted to ensure conformance to the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order).

5.1 Preservation, packing, packaging and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The Radio Set AN/URC-35( ) is intended for general purpose voice and CW communications, and may be used for surface ship, submarine, and fixed shore installations.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title number and date of this specification
b. Quantity of first article sample units required if other than specified in 4.3
c. Government furnished material (see 3.2)
d. Interchangeability requirement, if required (see 3.3)
e. Quantity of accessory items required (see 3.4.1)
f. Number of extender cables required (see 3.15.1.36)
g. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7.104.9(n) of the Armed Services Procurement Regulations (ASPR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7.104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirements	Applicable DID
4.5	Test Plans	DI-T-3709
4.5	Test Procedures	UDI-T-22710

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample. The first article should consist of at least one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Nomenclature. Nomenclature referred to herein and those assigned to the GFE units may be subject to change on any contract resulting in equipment produced in accordance with this specification.

6.5 Environmental inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.6 Standard definitions.

6.6.1 Standard dummy antenna. The standard dummy antenna is a device whose electrical constants are such as to simulate the impedance characteristics of antennas suitable for use with each equipment throughout its frequency range.

6.6.1.1 A device meeting the requirement of 6.6.1 for this equipment is a 50 ohm nonreactive resistance. This resistance includes the output resistance of the standard signal generator. If necessary, resistance shall be added external to the standard signal generator to produce a total of 50 ohms when added to the effective output resistance of the standard signal generator.

6.6.1.2 The intent of 6.6.1.1 is that the standard signal generator produce the equivalent of the signal voltage induced by an electromagnetic field in an antenna which has the constants that are simulated by the dummy antenna. Hence the voltage reading obtained from the standard signal generator is the open circuit voltage in series with the dummy antenna. This definition applies to all voltages specified relative to the receiver antenna input terminals. It should be noted that many standard signal generators do not indicate open circuit voltage, but rather the voltage which appears across a matched load. For these generators, the desired open circuit voltage is equal to twice the indicated output.

6.6.2 Standard output signal-plus-noise to noise ratio. Standard output signal-plus-noise to noise ratio (S+N/N) is 10 decibels (dB). This ratio shall be determined in terms of root mean square (rms) values.

6.6.2.1 For A1 and sideband reception modes, the output signal-plus-noise to noise ratio is the ratio between the audio output resulting from an unmodulated RF input signal producing the standard beat note frequency and the output due to noise when the RF input signal is removed.

6.6.2.2 For A2 and A3h modes of reception, the output signal-plus-noise to noise ratio is the ratio between the audio output resulting from an RF input signal modulated 30 percent at the standard modulation frequency and the output due to noise when the modulation is removed from the RF carrier.

6.6.3 Standard audio output. Standard audio output is a nominal six milliwatts dissipated into a 600 ohm nonreactive resistance across the audio frequency (AF) line output terminals.

6.6.4 Standard gain. Standard gain is that gain which produces standard output signal-plus-noise to noise ratio (see 6.6.2) with standard audio output.

6.6.5 Standard modulation frequency. Standard modulation frequency is 1000 Hz.

6.6.6 Standard beat note frequency. Standard beat note frequency for CW and single sideband reception is 1000 Hz.

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6.6.7 Sensitivity. Sensitivity of the receiver is that characteristic which determines the minimum strength of an input signal capable of producing a desired output signal-plus-noise to noise ratio.

6.6.7.1 Standard CW sensitivity. Standard CW (A1) sensitivity is the number of microvolts (open circuit) of unmodulated signal from a 50 ohm standard dummy antenna required to produce a standard output signal-plus-noise to noise ratio (see 6.6.2.1).

6.6.7.2 Standard single sideband sensitivity. Standard single sideband (A3j) sensitivity is the number of microvolts of unmodulated signal from a 50 ohm standard dummy antenna required to produce a standard output signal-plus-noise to noise ratio (see 6.6.2.1).

6.6.7.3 Standard AM sensitivity. Standard AM (A3h) sensitivity is the number of microvolts of signal, modulated 30 percent at standard modulation frequency, from a 50 ohm standard dummy antenna required to produce a standard output signal-plus-noise to noise ratio (see 6.6.2.2).

6.6.8 Selectivity. Selectivity of the receiver is that characteristic which determines the extent to which it is capable of discriminating between a desired RF signal and any undesired signals of frequencies outside the bandwidth required by the desired signal's information spectrum.

6.6.8.1 Front-end selectivity (antenna input to detector input). Front-end selectivity is determined primarily by the degree to which the IF amplifiers of the receiver are capable of rejecting undesired signals at frequencies immediately adjacent to the bandwidth required to pass a desired signal's information spectrum. Rejection of undesired signals is expressed in terms of decibels with respect to the peak of the desired signal passband ripple. Overall passband (antenna input to audio output) is determined by the combined effects of all frequency selective circuits in the entire signal path.

6.6.8.2 Overload selectivity. Overload selectivity (desensitization) is the degree to which the RF amplifiers of the receiver are capable of rejecting undesired signals at frequencies outside the passband of the RF circuitry. Protection of the receiver front end signal path circuits is required to avoid any decrease in a desired signal's signal-plus-noise to noise ratio by a strong undesired signal.

6.6.9 Spurious responses. In general, a spurious response in a receiver is the presence of any signal in the passband of the IF amplifiers which is not a function of the desired signal alone. There are several specific classes of spurious response:

a. Internally generated spurious outputs caused solely by the presence of the oscillators inherent to the receiver itself - also known as birdies
b. Signals which require the presence of a single strong signal external to the receiver - specific types are IF and image responses
c. Signals resulting from the intermodulation of two strong external signals in the first stages of the receiver prior to the IF amplifiers
d. The superposition of modulation from a single strong signal external to the receiver on a desired signal by a cross modulation process

6.6.9.1 Image rejection. Images are the undesired second order products of each conversion, resulting from a signal on the opposite side of the injection frequency from the desired response. Image rejection is expressed as the ratio of input signal amplitudes, in decibels, at the image and resonant frequencies, for standard output signal-plus-noise to noise ratio.

6.6.9.2 IF rejection. IF rejection is the degree to which the receiver is capable of rejecting signals having the frequency(s) of the IF amplifier(s). IF rejection is expressed as the ratio of input signal amplitudes, in decibels, at the IF and resonant frequencies, for standard output signal-plus-noise to noise ratio.

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6.6.10 Resonant overload. Resonant overload output is the output at which the overall gain (with no AGC) of the receiver, as conditioned for standard CW sensitivity, falls three dB with an increase of input carrier level.

6.6.11 Maximum undistorted output. Maximum undistorted power output is the output power level at which the total output harmonic content is two percent, in voltage, for an input carrier adjusted for the standard sideband measurement frequency (see 6.6.12) at the level of 1000 microvolts.

6.6.11.1 The total harmonic content is determined by measuring the residual voltages with the distortion analyzer after eliminating the fundamental of a sinusoidal modulating frequency.

6.6.12 Standard sideband measurement frequency. Standard sideband measurement frequency is 1000 Hz, that is 1000 Hz higher than the carrier for the upper sideband and 1000 Hz lower than the carrier for the lower sideband.

6.6.13 AGC time constants.

a. Attack time: The time from application of the large RF signal level until the audio output amplitude has stabilized within three dB of the final steady-state output

b. Hang time: The time from removal of the large RF signal level until the audio output voltage has risen to 10 percent of the steady state output due to the small signal

c. Decay time: The time after removal of the large RF signal, for the audio output voltage to rise from 10 percent to 63 percent of the final value due to the small signal

6.6.14 AGC threshold level. AGC threshold level is the RF input voltage for which the RF input/AF output characteristic departs from linearity by one dB.

6.6.15 Local input. The term local input in this specification indicates that control and modulating functions for the transmitter are being provided by the handset (or other appropriate device connected to the handset receptacle on the front panel) or by the hand key on the front panel.

6.6.16 Remote inputs. The term remote inputs in this specification indicates that control and modulating functions will be provided at the rear of the receiver-transmitter case.

6.6.17 Standard two-tone audio input. Standard two-tone audio input is defined as equal amplitude audio signals, 1000 Hz and 1625 Hz, with a level of 150 millivolts per tone at the remote input terminals.

6.6.18 Hum signals. Hum signals are any output products resulting from modulation at the primary power line frequency or power supply conversion frequencies or harmonics of these frequencies.

6.6.19 Standard single-tone audio input. Standard single-tone audio input is defined as a 1000 Hz tone, with a level of 150 millivolts at the remote input terminals.

6.6.20 Failure. Failure is defined as any malfunction or parameter deviation that prevents equipment from performing within the specified operational requirements.

6.7 Changes from previous issue. The extent of changes (deletions, additions, and so forth) precludes the annotation of the individual changes from the previous issue of this document.

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